Cancer Treatment Reviews 139 (2025) 102981

Contents lists available at ScienceDirect

Cancer Treatment Reviews

journal homepage: www.elsevier.com/locate/ctrv

Systematic or Meta-analysis Studies

Surgical tumor volume reduction in patients with brain metastases: A systematic review and *meta*-analysis

Vittorio Stumpo [a,1], Alessandro Carretta [a,b,1], Jacopo Bellomo [a], Luis Padevit [a], Victor Staartjes [a], Nicolai Maldaner [a], Penelope Coker [a], Jorn Fierstra [a], Michael Weller [c], Emilie Le Rhun [d], Marcus Czabanka [e], Oliver Bozinov [f], Luca Regli [a], Marian Christoph Neidert [f], Carlo Serra [a], Stefanos Voglis [a,*]

[a] Department of Neurosurgery, Clinical Neuroscience Center, University Hospital and University of Zurich, Zurich, Switzerland
[b] Department of Biomedical and Neuromotor Sciences (DIBINEM), University of Bologna, Bologna, Italy
[c] Department of Neurology, Clinical Neuroscience Center, University Hospital and University of Zurich, Zurich, Switzerland
[d] Department of Medical Oncology and Hematology, University Hospital and University of Zurich, Zurich, Switzerland
[e] Department of Neurosurgery, University Hospital Frankfurt, Frankfurt, Germany
[f] Department of Neurosurgery, Cantonal Hospital St. Gallen, St. Gallen, Switzerland

ARTICLE INFO

Keywords:
Brain metastases
Extent of resection
Residual volume
Gross total resection
Meta-analysis

ABSTRACT

Background: Microsurgical resection of brain metastases (BM) has traditionally been a mainstay of local control for large or symptomatic lesions. Maximal tumor burden reduction remains controversial in the multidisciplinary management of BM patients and needs to be re-evaluated in view of new systemic treatment options. We conducted a systematic review and *meta*-analysis to evaluate the role of extent of resection (EOR)/residual volume (RV) for progression-free (PFS) and overall survival (OS) in patients with BM.
Methods: A systematic review was performed according to PRISMA guidelines and included studies' quality was assessed with the Grading *of Recommendations,* Assessment*, Development, and* Evaluations (GRADE) tool. Study characteristics were tabulated and critically reviewed. Results from Cox-regression models and log-rank tests for the association of gross total resection (GTR) versus subtotal resection (STR) with PFS and OS were extracted to perform separate random-effects *meta*-analyses.
Results: Thirty-nine articles were included, all of them being retrospective and all but 3 monocentric. Most studies included BM from heterogenous primary tumors, with 9 focusing on BM from a single primary. Systemic therapies were variably reported and only 2 studies reported on the use of steroids. Twenty-one studies showed a significant association of EOR/RV with improved OS. Meta-analysis of studies reporting multivariable Cox-regression models ($n = 11$) showed a significant association of GTR with longer OS (HR 0.67, 95 %CI 0.56–0.81, $p < 0.001$).
Conclusion: Although in several studies higher EOR/lower RV was associated with improved OS, evidence consists of heterogeneous cohorts and rarely includes primary tumor-specific systemic therapies or relevant confounding covariates. New studies are needed to elucidate the role of microsurgical tumor burden reduction in BM patients in the era of targeted or immune modulatory therapies.

Introduction

Brain metastases (BM) are diagnosed in around 20 % of cancer patients, and are most commonly associated with lung cancer (20–56 % of patients), breast cancer (5–20 %) and melanoma (7–16 %), however virtually any tumor can metastasize to the brain[1]. The multidisciplinary treatment of BM involves surgical resection, radiotherapy (RT) and systemic therapies and depends heavily on the primary tumor type, the molecular characteristics of the primary tumor and the BM, the anatomical location of the BM as well as previous therapies already

* Corresponding author.
E-mail address: Stefanos.Voglis@usz.ch (S. Voglis).
[1] These authors contributed equally.

https://doi.org/10.1016/j.ctrv.2025.102981
Received 6 March 2025; Received in revised form 10 June 2025; Accepted 15 June 2025
Available online 16 June 2025
0305-7372/© 2025 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

administered, overall prognosis and performance status of the patient [2]. The indications for surgical resection of BM include single BM, diagnostic uncertainty in cases where no histological diagnosis of a primary tumor is available, symptomatic lesions when other options may be beneficial for the patient [2,3].

In patients with diffuse gliomas, there is substantial evidence demonstrating the association of reduction in intracranial tumor burden (measured by either extent of resection, EOR, or residual tumor volume, RV) with improved progression-free (PFS) and overall survival (OS), resulting in implications for clinical practice[4]. However, analogous evidence in the context of BM remains scarce. Compared to gliomas for which there are often clear guidelines for therapies beyond surgery[5,6], the multidisciplinary treatment of BMs includes a continuously increasing number of systemic and local therapies, such as stereotactic radiosurgery (SRS) which makes assessing the role of surgical resection more challenging[7].

In light of new systemic immunotherapeutic and targeted treatments for BM patients, the role of surgical resection, and more importantly the role of EOR/RV as a surrogate for central nervous system disease burden for survival outcomes remains scarcely investigated[8,9]. Conversely, the evolution of microsurgical techniques in recent years made it possible to resect BM with very low morbidity, even in anatomical locations once deemed surgically challenging[10,11]. Consequently, the role of surgical tumor burden reduction needs to reassessed in terms of oncological outcomes, weighting the potential benefits of resection against the patient's clinical status – a key prognostic factor not only for surgical outcome but also for overall oncological prognosis.

Therefore, in the present study, we systematically reviewed and critically discuss the available literature to evaluate the role of surgical tumor burden reduction in terms of EOR and/or RV for survival outcome in patients with BM.

Methods

Overview

The systematic review was performed according to Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) protocol[12] and registered in PROSPERO (https://www.crd.york.ac.uk/PROSPERO/view/CRD42024548670). Literature published until October 2022 was searched on PubMed, Scopus, Web of Science and Google Scholar databases. A repeat search was performed in January 2024.

Search strategy

Detailed search strings are shown in Supplementary Table 1. After duplicate removal, titles and abstracts were first screened, and for the papers deemed appropriate, the full text was obtained and reviewed for appropriateness and extraction of data. The articles' reference lists were additionally screened to identify any other relevant studies.

Study selection

Study selection was performed based on the following criteria: Prospective or retrospective studies reporting surgical resection of one or multiple BM on human subjects reporting either GTR vs STR or EOR or RV as well as outcome data (PFS, OS). PICO criteria were as follows: (P), adult patients aged 18 years or more; (I) GTR; (C), STR; (O), OS and PFS. Case reports, small case series, reviews, articles not in English language and all those lacking adequate data reporting to compare the different EORs or RVs in relation to PFS/OS were excluded. To get an unbiased overview of the available literature, no time limit of published studies was applied. The PRISMA flow diagram is shown in Supplementary Fig. 1.

Data extraction and quality assessment

The individual steps of title and abstract screening, full-text review and data extraction were performed independently by two reviewers (VS and AC), and disagreements at any stage were resolved by discussion and consensus. Methodological quality of included studies was graded using the *Grading* of *Recommendations Assessment*, *Development* and *Evaluation* (GRADE) tool[13]. Data from the included studies were extracted, tabulated, and analyzed. Collected variables included first author, publication year, inclusion timeframe, study design, number of patients, sex, age, primary tumor histology, number of BM in included study cohorts, primary or recurrent setting, treatment of solitary or multiple BM, pre- and postoperative tumor volume, pre- and postoperative neurological status, information on extracranial status and number of extracranial metastases, complications, use of steroids, systemic therapies (including chemotherapy, targeted therapy or immunotherapy), local treatment including SRS or RT use, extent of surgical resection (GTR/STR/EOR/RV), survival outcome data (PFS/OS), statistical analysis method used to assess association of EOR with outcome (Cox-regression model, log-rank statistics, others), and when available hazard ratios (HR), 95 % confidence interval (CI) and p-values.

Data analysis

Extracted data were tabulated for visual overview. Summary plots were produced using R (version 4.4.1)[14] with R Studio (version 2024.09.0)[14] and open-source libraries (meta and survival package). For studies using Cox-regression models, HRs, and 95 %CIs for GTR versus STR were tabulated. When multivariate Cox-regression models were available, these datapoints were collected. For studies only using univariate analysis, the univariate HRs were extracted. When only Kaplan-Meier (KM) curves or log-rank p-values were reported, we estimated the HRs using the methodology reported in Supplementary File 1 [15–19].

Based on anticipated different methodologies in reporting survival data, we performed *meta*-analyses on subgroups based on the statistical methodologies used in the respective studies[19,20]: a separate *meta*-analysis was run for each to increase statistical power: *meta*-analysis of studies reporting results of multivariate Cox-regression models, *meta*-analysis of studies reporting results of univariate Cox-regression model (for studies reporting both uni- and multivariate Cox-regression model the results of univariate analysis were included in this second subgroup, also if already included in the *meta*-analysis of multivariate studies) and *meta*-analysis of studies reporting results of log-rank tests.

Random effects *meta*-analyses were performed as described in Supplementary file 1 [19,21,22]. Pooled HRs with 95 %CIs were calculated, with HR < 1 indicating a survival benefit for GTR versus STR. To assess the potential for publication bias, we performed Egger's test and visually inspected a funnel plot[23,24]. A two-sided significance level of 0.05 was chosen for statistical analyses. Leave-one-out analysis was performed to assess the impact of each individual study on the overall results[25].

Cohort subgroup meta-analyses' feasibility was evaluated based on the following categories: time of publication (articles published in the last 10 years), primary tumor histology, first operation, operation for a recurrent tumor, studies adopting same RV thresholds and single versus multiple lesions resection in multiple metastases patients. A subgroup random-effect *meta*-analysis was judged feasible if at least three studies were present in each of these categories and conducted according to the same methodology as reported. In anticipation of heterogeneous statistical reporting and given the exploratory nature of these subgroup analyses, pooling of studies with different statistical designs was done.

Table 1
Studies overview, demographics, primary tumors, number of metastases and study grading.

Authors	Year of publication	Time frame of inclusion	Study design	Patients (n)	Sex F/M (%)	Age	Primary Tumor	Metastases (n)	GRADE score
Sauvageot et al.	**2024**	2015–2021	RS	184	F: 77/184 (41.9 %) M: 107/184 (58.1 %)	Mean: 60.0 (± 9.9)	LC: 184	1: 108/184 (58.7 %) 2: 36/184 (19.6 %) 3: 22/184 (11.9 %) >3: 18/184 (9.8 %)	Moderate
Grossenbacher et al	**2023**	2004 – 2014	RS	285	F: 136/285 (52.3 %)M: 149/285 (47.7 %)	Median: 62.2 (range: 29.9 – 86.6	LC: 118, Mel: 50, BC: 29, UGT: 11, GIT: 38, Other: 39	1–4: 262/285 (91.9 %) >4: 23/285 (8.1 %)	Moderate
Baumgart et al	**2023**	2007–2020	RS	101 (> 75 yrs)	F: 44/101 (44.0 %) M: 57/101 (56.0 %)	Median:78 (IQR 76–81)	LC: 23, Mel: 22, BC: 11, UGT: 17, GIT: 9, Other:19	1: 62/101 (61 %) 2: 16/101 (16 %) 3: 13/101 (13 %) >3: 10/101 (10 %)	Moderate
Lin et al	**2023**	2007–2022	RS	219 (94 surgery group)	F: 121/219 (55.3 %) M: 98/219 (44.7 %)	Median: 60.0 (IQR 52–69)	NA	1: 82/219 (37.4 %) >1: 137/219 (62.6 %) Operated 1: 56/95 (59.0 %) 2: 19/95 (20.0 %) 3: 10/95 (10.5 %) >3: 10/95 (10.5 %)	Low (Failure to adequately control confounding)
Voglis et al	**2022**	2013—2021	RS	121	F: 41/121 (34.0 %) M: 80/121 (66.0 %)	Mean: 59.93 (± 14.15)	Mel: 121	1: 52/121 (43 %) <10: 51/121 (42 %) >10: 18/121 (15 %)	Moderate
Henderson et al	**2022**	2014—2021	RS	112	F: 68/112 (61.0 %) M: 44/112 (39 %)	Median: 65 (range: 24–84)	LC: 65, Mel: 7, BC: 14, UGT: 13, GIT: 12, Other: 1	1: 81/112 (72 %) >1: 29/112 (26 %)	Low (Failure to adequately control confounding factors)
Aftahy et al	**2022**	2007—2020	RS	704	F: 350/704 (49.7) M: 354/704 (50.3)	Median: 64.0 (range: 18–93	NA	1: 372/704 (52.8 %) 2: 122/704 (17.3 %) 3: 142/704 (20.2 %) >3: 68/704 (9.7 %)	Low (Failure to adequately control confounding factors)
Sander et al	**2022**	2012—2020	RS	25	F: 6/25 (24.0 %) M: 19/25 (76.0 %)	Mean: 61.88 (range 30–81)	GIT: 25	1: 14/25 (56.0 %) 2: 3/25 (12.0 %) 3: 6/25 (24.0 %) >3: 2/25 (8.0 %)	Very low (small study population and small magnitude of effect)
Winther et al	**2022**	2011—2018	RS	373	F: 195/373 (52.0 %) M: 178/373 (48.0 %)	Median: 63 (range: 18–89)	LC: 134, Mel: 89, BC: 26, UGT: 19, GIT: 39, Other: 66	1: 374 (100 %)	Moderate
Shah et al	**2021**	2009–2019	RS	84	F: 50 (60.0 %) M: 34 (40.0 %)	Mean: 63 (±11)	LC: 84	1: 36/84 (43.0 %) 2–3: 24/84 (29.0 %) >3: 24/84 (29.0 %)	Low (small study cohort)
Koo et al	**2021**	2004—2020	RS	124	F: 65/125 (52.4 %) M: 59/125 (47.6 %)	Median: 59 (range: 28–78)	LC: 50, Mel: 3, BC: 34, UGT: 16, GIT: 16, Other: 5	1: 64/124 (51.6 %) >1: 60/124 (48.4 %)	Low (small study cohort)

(*continued on next page*)

Table 1 (*continued*)

Authors	Year of publication	Time frame of inclusion	Study design	Patients (n)	Sex F/M (%)	Age	Primary Tumor	Metastases (n)	GRADE score
Prabhu et al	**2021**	NA	RS MC	242	F: 138/242 (57.0 %) M: 104/242 (43.0 %)	Median: 59.8 (IQR 53–68.2)	LC: 105, Mel: 31, BC: 49, UGT: 19, GIT: 17, Other: 21	1: 151/242 (62.4 %) 2: 46/242 (19 %) 3: 27/242 (11.2 %) 4: 11/242 (4.5 %) ≥5: 7/242 (2.9 %)	Moderate
Jünger (Cancers) et al	**2021**	2010—2019	RS	197	F: 109/197 (55.3 %) M: 88/197 (44.7.0 %)	Median: 62 (range: 19–87)	LC: 92, Mel: 20, BC: 27, UGT: NA, GIT: 33, Other: 24	NA	Moderate
Jünger (JNS) et al	**2021**	2008–2020	RS	216	F: 95/216 (44.0>%) M: 121/216 (56.0 %)	Median: 62 (range: 38–87)	LC: 216	1: 129/216 (28.7 %) 2–3: 64/216 (29.6 %) >4: 23/216 (10.6 %)	Moderate
McHugh et al	**2020**	2005—2017	RS	110	F: 41/110 (37.0 %) M: 69/110 (63.0 %)	Mean: 59.9 (range: 22.3–88)	Mel: 110	1: 73/110 (66.4 %) 2: 19/110 (17.3 %) 3: 9/110 (8.15 %) >3: 9/110 (8.15 %)	Moderate
Eitz et al	**2020**	2003–2019	RS MC	558	F: 301/558 (53.9 %) M: 257/558 (46.1 %)	Mean: 61 (± 0.50)	LC: 194, Mel: 89, BC: 98, UGT: 47, GIT: 65, Other: 88	1: 373/558 (64.2 %) 2–3: 180/558 (31.0 %) >4: 28/558 (4.8 %)	Low (Failure to adequately control confounding factors)
Schödel et al	**2020**	2010–2019	RS BC	750	F: 379/750 (50.5 %) M: 371/750 (49.5 %)	Median: 61 (range: 19–87)	LC: 318, Mel: 114, BC: 116, UGT: 24, GIT: 72, Other: 106	1: 462/750 (61.6 %) 2–3: 185/750 (24.7) >3: 103/750 (13.7 %)	Moderate
Olesrud et al	**2019**	2014—2016	RS	68	F: 46/68 (68.0 %) M: 22/68 (32.0 %)	Median: 69; Mean: 65 (range: 42–85)	LC: 33, Mel: 7, BC: 8, UGT: 11, GIT: 9	1: 55/68 (81 %) 2: 10/68 (15 %) 3: 3/68 (4 %)	Moderate
Calluaud et al	**2019**	2006—2017	RS	120	F: 49/120 (40.8 %) M: 71/120 (59.1 %)	Median 61.9 (range: 55.3–88.9)	LC: 67, BC: 14, UGT: 9, GIT: 18, Other:6	Mean: 1.8/patient	Moderate
She et al	**2019**	2001—2014	RS	122	F: 49/122 (40.2 %) M: 73/122 (59.8 %)	Median: 56 (range: 31–81)	LC: 122	1: 50/122 (41 %) ≥ 2: 72/122 (59 %)	Low (failure to adequately control confounding factors)
Sivasanker et al	**2018**	2007–2015	RS	124	F: 71/124 (57.3 %) M: 53/124 (42.7 %)	Median: 53	LC: 25, BC: 45, UGT: 24 GIT: 9, Other:21	1: 119/124 (96 %) ≥ 2: 5/124 (4 %)	Low (failure to adequately control confounding factors)
Antuna et al	**2018**	2002–2015	RS	71	F: 26/71 (36.6 %) M: 45/71 (63.4 %)	Mean: 58.8 (range: 43–82)	LC: 71	1: 57/71 (80.3 %) 2–3: 14/71 (19.7 %)	Low (small study cohort)
Kerschbaumer et al	**2017**	1999—2013	RS	143	76/143 (53.1) 67/143 (46.9)	Range: 34–85	LC: 59, Mel: 19, BC: 19, UGT: 11, GIT: 13, Other: 22	NA	Low (poor quality of outcome reported)
Kamp et al	**2017**	2010–2017	RS	36	F: 18/36 (50.0 %) M: 18/36 (50.0 %)	Mean: 58.6 (± 1.8)	LC: 23, Mel: 3, BC: 2, UGT: 2, GIT: 5, Other: 1	1: 35/36 (97 %) 2: 1/36 (3 %)	Very low (small study population, failure to adequately control confounding factors)
Rava et al	**2016**	2002—2010	RS	87	F: 51/87 (59 %) M: 36/87 (41 %)	Median: 59 (range: 37–84)	LC: 44, Mel: 8, BC: 9, UGT: 7, GIT: NA, Other: 19	1: 50/87 (58 %) >1: 37/87 (42 %)	Moderate

(*continued on next page*)

Table 1 (*continued*)

Authors	Year of publication	Time frame of inclusion	Study design	Patients (n)	Sex F/M (%)	Age	Primary Tumor	Metastases (n)	GRADE score
Gazzeri et al	**2014**	2000—2008	RS	57	F: 30/57 (52.0 %) M: 27/57 (48.0 %)	Mean: 53.98 (range: 32–74)	LC: 14, Mel: 11, BC: 17, UGT: 8, GIT: NA, Other: 7	1: 22/57 (38.6 %) >1: 35/57 (61.4 %)	Low (small magnitude of effect)
Lee et al	**2013**	1995—2011	RS	157	F: 75/157 (47.8 %) M: 82/157 (52.2 %)	Mean: 53.7 (range: 23–78)	LC: 72, Mel: 8, BC: 21, UGT: 19, GIT: 16, Other: 21	1: 96/157 (61.1 %) 2: 61/157 (38.9 %)	Moderate
Schackert, Schmiedel et al	**2013**	2000–2008	RS	127	F: 48/127 (37.8 %) M: 79/127 (62.2 %)	Median: 67 (range: 19–81)	LC: 44, Mel: 22, BC: 15, UGT: 15, GIT: 16, Other: 15	1–4: 100/127 (78.7 %) >4: 27/127 (21.3 %)	Low (failure to adequately control confounding factors)
Schackert, Lindner et al	**2013**	1993–2008	RS	67	F: 34/67 (50.7 %) M: 33/67 (49.3 %)	Median: 59 (range: 30–75)	LC: 10, Mel: 12, BC: 11, UGT: 13, GIT: 12, Other: 9	1: 41/67 (61.2 %) >1: 26/67 (38.8 %)	Very low (small study population, failure to adequately control confounding factors)
Hong et al	**2013**	2001–2007	RS	51	F: 15/51 (29.4 %) M: 36/51 (70.6 %)	Mean: 55.9 (range: 29–73)	LC: 51	< 3: 10/51 (26.3 %) 3–4: 19/51 (50.0 %) >5: 9/51 (23.7 %)	Very low (small study population, risk of bias)
Schödel et al	**2013**	NA	RS	206	F: 84/206 (40.78 %) M: 122/206 (59.2 %)	Mean: 61.6 (range: 23.4–83.9)	LC: 70, Mel: 30, BC: 28, UGT: 27, GIT: 20, Other: 31	Solitary 61 (29.7 %) Single 60 (29.1 %) Multiple 85 (41.2 %)	Low (failure to adequately control confounding factors)
Auslands et al	**2012**	2005—2010	RS	36	F: 18/36 (50.0 %) M: 18/36 (50.0 %)	Mean: 58.47 (± 10.98, range: 35–82)	LC: 2, Mel: 15, BC: 9, UGT:3, GIT: 3, Other: 4	2–3: 27/36 (75 %) 4–6: 9/36 (25 %)	Very low (small study population, failure to adequately control confounding factors)
Limbrick et al	**2009**	1998—2003	RS	15	F: 10/15 (66.7 %) M: 5/15 (33.3 %)	Median: 56.8 (range: 41–85)	LC: 6, BC: 4, UGT: 4 GIT: 1, Other: NA	1: 12/15 (80 %) 2: 3/15 (20 %)	Very low (small study population, failure to adequately control confounding factors)
Yoo et al	**2009**	2001–2007	RS	94	F: 27/94 (28.7 %) M: 67/94 (71.3 %)	Mean: 55 (±12)	LC: 60, Mel: NA, BC: 6, UGT: 9, GIT: 16, Other: 3	NA	Very low (small study population, failure to adequately control confounding factors, poor quality of outcome reported)
Rades et al	**2007**	1995—2005	RS	201	F: 95/201 (47.3 %) M: 106/201 (52.7 %)	<62: 111/201 (55 %) >63: 90/201 (45 %)	LC: 82, Mel: NA, BC: 43 UGT: NA, GIT: 23, Other: 53	NA	Moderate
Tendulkar et al	**2006**	1984–2004	RS	271	NA	<65: 172/271 (63 %) >65: 99/271 (37 %)	LC: 108, Other: 163	NA	Low (failure to adequately control confounding factors)
Paek et al	**2005**	1995–2002	RS	208	F: 105/208 (50.5 %) M: 103/208 (49.5 %)	Median: 59 (range: 31–82)	LC: 105, Mel: 18, BC: 28, UGT: 13, GIT: 12, Other: 40	1: 132/208 (63.5 %) >1: 76/208 (36.5 %)	Moderate
Stark et al	**2005**	1994–2001	RS	177	NA	Median: 59, Mean: 58.5 (range: 32—86)	LC: 96, Mel: NA, BC: 35, UGT: 17, GIT: 17, Other: 12	1: 78/177 (44.1 %) >1: 96/177 (54.2 %) 1–3: 161/177 (91 %) >3: 13/177 (7.3 %)	Moderate
Bindal et al	**1993**	1984–1992	RS	82	F: 43/82 (52.4 %) M: 39/82 (47.6 %)	Median multiple (Group A and B): 52 (range 23–76). Median single (Group C): 56 (range: 24–71)	LC: 10, Mel: 37, BC: 16, UGT: 4, GIT: 5, Other: 10	1: 26/82 (31.7 %) >1: 56/82 (68.3 %)	Very low (small study population, failure to adequately control confounding factors)

Abbreviations. BC: breast cancer; BM: brain metastasis; CUP: carcinoma of unknown primary; CRC: colorectal cancer; EOR: extent of resection; GIT: gastrointestinal tract; GUC: genito-urinary carcinoma; GTR: gross total resection; HR: hazard ratio; HSRS: hypofractioned stereotactic radiosurgery; ICI: immune checkpoint inhibitor; LC: lung cancer; LBRT: local – brain radiotherapy; Mel: melanoma; NSCLC: non-small cell lung cancer; OC: ovarian cancer; RPA: recursive partitioning analysis; RV: residual volume; RT: radiotherapy (not specified); RS: retrospective study; SCC: squamous cell carcinoma: SRS: stereotactic radiosurgery; STR: subtotal resection; WBRT: whole – brain radiotherapy; yrs: years.

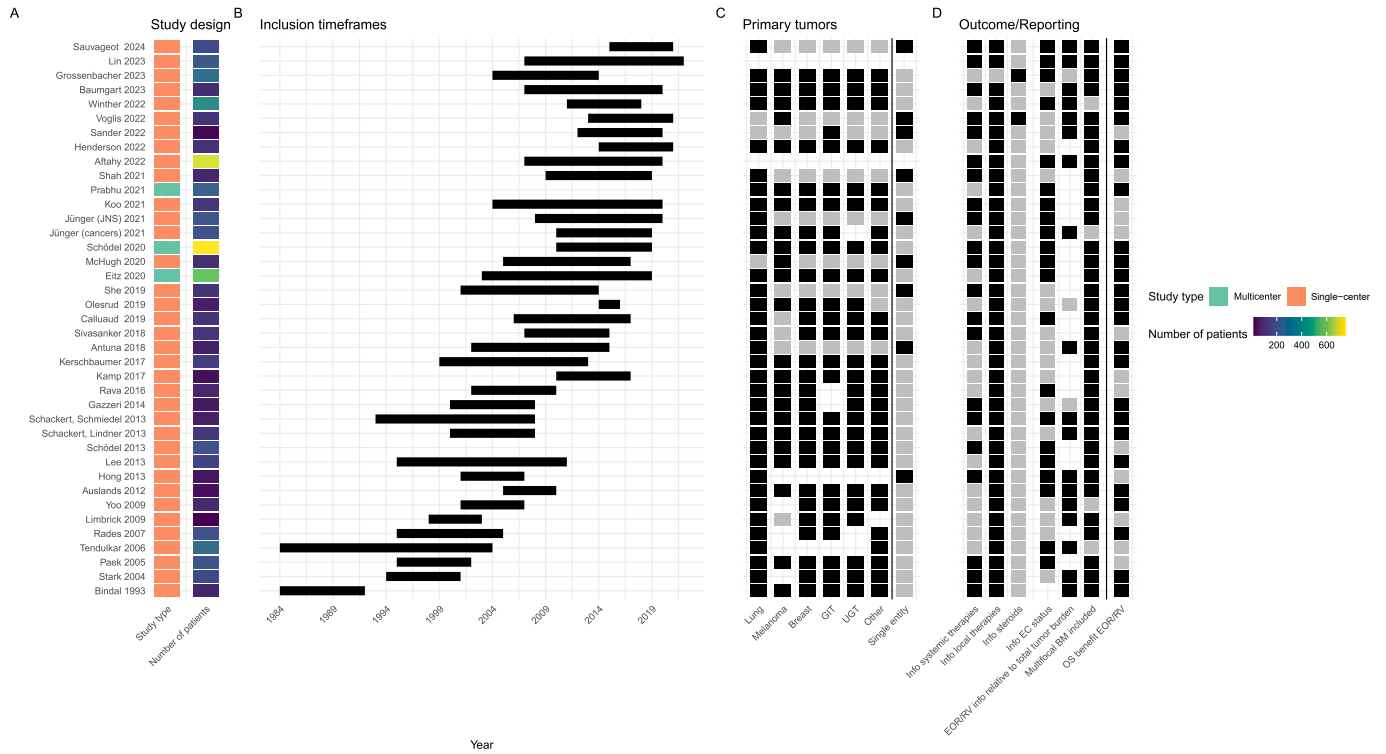

Fig. 1. Overview of included studies: (A) First author, year, study type and number of patients included in the respective study. (B) Timeframe of patient inclusion. (C) Primary tumor types included in the study and whether only single entities were included. (D) Covariate and outcome reporting. GIT gastrointestinal tumors; UGT urogenital tumors; EC extracranial; EOR extent of resection; RV residual volume; OS overall survival.

Results

Literature search

The literature search yielded a total of 4260 articles. After the removal of duplicates (n = 2381), abstracts of 1879 articles were screened; 1587 were found not to be pertinent to the current investigation and were excluded and 31 could not be retrieved. After full-text review of the remaining 261 articles, 20[8,26–44] were included for qualitative analysis (reasons for exclusion during full-text screening are reported in Supplementary Fig. 1). Three articles [9,45,46] were identified after a repeat search performed in January 2024 and were added to the final count (n = 23). An additional screening of references of the included studies (n = 766) was performed

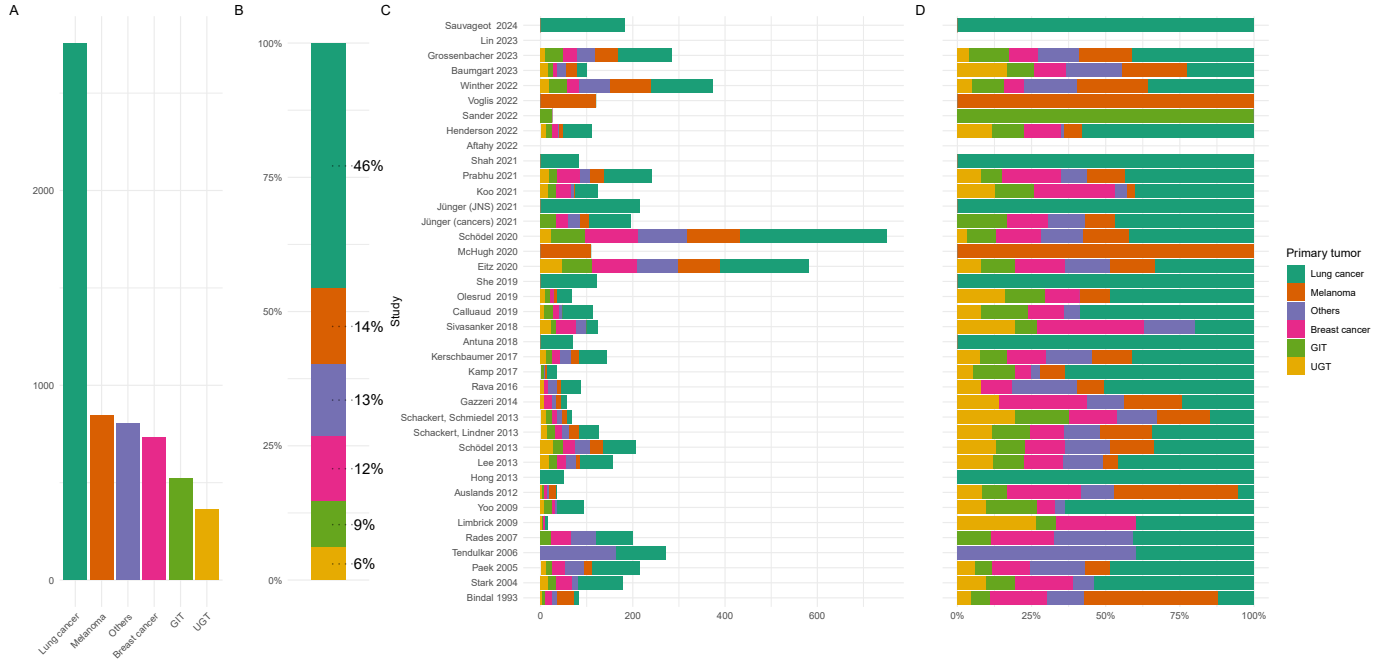

Fig. 2. Primary tumor numbers: Overview of different primary tumor numbers in all included studies with absolute (A) and relative (B) numbers. Absolute (C) and relative (D) primary tumor numbers per study. GIT gastrointestinal tumors; UGT urogenital tumors;

Table 2
Overview of systemic, local and surgical treatment reported in the included studies and summary of main findings concerning role of surgical resection for overall survival and progression-free survival.

Author	N. Patients	Systemic treatment	Local treatment	OS/PFS	GTR/STR/ EOR/RV	Total IC TB	EOR/RV association with OS/PFS?	Notes
Sauvageot et al	184	154 Ch (83.7 %) 63 IT (34.2 %) (27 incl. adjuvant IT, i.e. 14.7 %) 13 TT (7.1 %)	161 RT- WBRT: 26 (16.2 %), HSRT:123 (76.4 %), SRS:1 (0.6 %), SRS + HSRT:11(6.8 %)	Median OS 19.2 mos (95 % CI 13.2 to 24.0) Median PFS 8.4 mos (95 % CI, 7.2 to 12.0)	T 181- GTR: 137 (74.5 %), STR: 44 (24.3 %), 3 pts missing	Yes	OS: **Yes***(Only UV), RV > 0.5 cm3; MV Cox regression analysis – no association. UV Cox regression, p = 0.0003, HR 2.34 (95 % CI, 1.52 to 3.58); PFS: **No**, RV > 0.5 cm3 UV Cox regression analysis, p = 0.724. HR 1.07 (95 % CI, 0.73–1.56)	RV significant for OS only at UV analysis. MV Cox regression analysis: age < 65 yrs, single BM, adjuvant brain RT, adjuvant TT, and total pre-op TB < 7 cm3 – favorable prognostic factors of OS. No association of RV w/ PFS.
Baumgart et al	101	23 Ch (23.0 %) 14 IT (14.0 %)	28 WBRT (28.0 %) 2 SRS (2.0 %) 34 HSRT (34.0 %)	Median OS [IQR] 7 mos [2,11]	RV = 0: 52(51.5 %), RV > 0: 49 (48.5 %) Only 1 met (n = 62) 48 GTR (77 %), 1 4 STR (23 %)	Yes	OS: **Yes**, RV > 0 cm3: MV Cox regression, p = 0.010, HR 1.90 (95 % CI, 11.898 (1.162–3.099). UV Cox regression, p = 0.029: HR (1.025, 95 % CI 1.002–1.047. Patients w/ RTB = 0 – ↑ survival rates (p = 0.007) and ↑ median OS: 12 (IQR 5–19) vs 5 (IQR 3–7) mos (KM), Median [IQR] OS 12 [5,19] for RV = 0 cm3 vs 5 [3,7] for RV > 0 cm3	Only pts > 75 yo. MV Cox regression analysis: no post-op RV only favorable prognostic factor for OS. Pts w/ multiple BMs and GTR of targeted BM had ↑ OS vs pts with STR of target BM (p = 0.042)
Lin et al	219	100 (45.7 %) nos	154 (70.3 %) nos	Median OS [IQR] 6 mos [2,10]	RTB = 0: 55 (58.5 %) RTB > 0: 40 (41.5 %)	Yes	OS: **Yes**, post-op RV: MV Cox regression, p = 0.007, HR 1.03 (95 % CI, 1.01 to 1.05). UV: NA.	MV Cox regression: low post-op TB, GTR independent favorable prognostic factors for OS.
Grossenbacher et al	285	NA	NA	Median OS 14 mos (95 % CI 11.9–16.1)	GTR: 136 (47.7 %) STR: 149 (52.3 %)	No	OS: **Yes,** RV > 0: MV Cox regression, p = 0.007; HR 1.6 (95 %CI 1.1–2.3). UV: NA.	MV Cox regression, measurable RV, age > 60, KPS < 80, CSF infection, nEC mets sites and high dose dexamethasone associated w/ decreased OS
Voglis et al	121	ICI-naive with postoperative ICI	WBRT: 20 (16.5 %), SRS: 66 (55.0 %), SRS + WBRT: 6 (5.0 %), None: 29 (24.0 %)	Median OS 19.75 mos	NA	Yes	OS: **Yes**, post-op RV > 3.5 cm: MV Cox regression, p = 0.027, HR 1.4 (95 % CI, 1.04 to 1.88). UV: NA.	MV Cox regression: post-op ICI, low RV independent favorable prognostic factors for OS. Only melanoma BMs included.
Henderson et al	112	NA	27 WBRT (24.0 %) 18 SRS (16.0 %) 1 SRS + WBRT (1.0 %) 71 none	Median OS 7.6 mos	GTR: 46 (41.0 %), STR: 26 (23.0 %), Debulking: 9 (8.0 %), Unknown: 31 (28.0 %)	NA	OS: **Yes**, EOR: KM analysis, log-rank p = 0.022. Median OS GTR vs STR vs Debulking: 11.8 vs 5.7 vs 7 mos.	Only KM log-rank Mantel-Cox analysis. GTR and younger age favorable prognostic factors for OS.
Aftahy et al	704	Ch: 301 (42.8 %), IT: 76 (10.8 %)	208 WBRT (29.5 %) 26 SRS (3.7 %) 231 HSRS (32.8 %)	Median OS [IQR] 6 [2,18] mos	NA	Yes	OS: **Yes**, post-op TB: MV Cox regression, p = 0.0036, HR 1.02 (95 % CI, 1.01 to 1.03); no UV: NA.	MV Cox regression analysis: high pre-op KPS, younger age, low pre-op TB, low post-op RV independent favorable prognostic factors for OS.
Sander et al	25	Ch and IT (nos)	WBRT: 6 (24.0 %) SRS 11: (66.0 %)	Median OS 3 mos (min–max 0–18)	NA	Yes	OS: **No**, post-op RV > 0: UV Cox regression, p = 0.948 HR 1.03 (95 %CI, 0.41–2.60). MV Cox regression analysis not significant.	UV Cox regression analysis: high KPS, good post-op neurological status, WBRT independent favorable prognostic factors for OS. None significant at MV analysis.
Winther et al	373	NA	WBRT: 147 (39.0 %), SRS: 58 (16.0 %), PBRT: 19 (5.0 %), SRS + WBR: 4 (1.0 %),	Median OS 11 mos	NA	Yes	OS: **Yes**, GTR: MV Cox regression, p = 0.003, HR 0.66 (95 % CI, 0.51 to 0.87); UV Cox regression, p = 0.001, HR 0.65 (95 % CI,	MV Cox regression: GTR, female sex, younger age, EC metastases, low ECOG independent favorable

(*continued on next page*)

Table 2 (*continued*)

Author	N. Patients	Systemic treatment	Local treatment	OS/PFS	GTR/STR/ EOR/RV	Total IC TB	EOR/RV association with OS/PFS?	Notes
			None: 114 (31.0 %), Unknown: 31 (8.0 %)				0.53–0.85) GTR vs STR: 13 vs 8 mos	prognostic factors for OS. Subgroup analyses for lung and melanoma.
Shah et al	84	TKI	Pre-op: SRS 12 (14.0 %), WBRT 6 (7.0 %), None 66 (79.0 %). Post-op: SRS 45 (55.0 %), WBRT 14 (17.0 %), None 24 (29.0 %)	Median OS 9.7 mos (95 % CI: 7.17–25.05)	GTR: 70 (83.3 %) STR: 14 (26.7 %)	NA	OS: **No**, STR: UV Cox-regression– p = 0.2, HR 1.64 (95 %CI, 0.76 –3.53).	No association at UV or MV of GTR w/ ↑ OS. MV Cox regression: GPA score ≤ 1 significantly associated w/ decreased OS. Post-op SRS, resected BM diameter < 3 cm significantly associated with prolonged OS.
Koo et al	124	TT: 15 (12.1 %)	WBRT: 24 (19.3 %), LBRT: 26 (21.0 %), SRS: 37 (29.8 %), None: 37 (29.8 %)	Median OS 21.1 mos (95 % CI: 15.8 to 26.4)	GTR: 109 (87.9 %) STR: 15 (12.1 %)	NA	OS: **No**, GTR: UV Cox regression, p = 0.996, HR 1.0 (95 %CI 0.48–2.08) LR: **Yes** GTR: MV Cox regression, p = 0.014 HR 0.29 (95 %CI 0.11–0.78).	MV Cox regression: KPS > 90, TT, systemic control, single BM independent favorable prognostic factors.
Prabhu et al	242	NA	Pre-op SRS: 242 (100.0 %)	Median OS 16.9 mos (95 % CI, 13 to 20.8)	NA	NA	OS: **Yes**, GTR: MV Cox regression, p = 0.005, HR 0.43 (95 %CI 0.24–0.79). UV not reported. LR: Yes, GTR: UV Cox analysis	MV Cox regression: GTR, younger age, synchronous BM number, melanoma, GPA independent favorable prognostic factors for OS.
Jünger et al (Cancers)	197	NA	WBRT: 47 (23.9 %) LBRT: 142 (72.1 %) SRS: 6 (3.0 %)	Median OS 12 (95 % CI, 8.6 to 154)	GTR: 123 (62.3 %) STR: 74 (37.7 %)	Yes	OS: **No**, GTR; KM analysis, log-rank p = 0.759. No median OS GTR vs STR reported. PFS: NO, GTR; KM analysis, log-rank = 0.1395	MV Cox regression: systemic Tx, controlled systemic disease independent favorable prognostic factors for OS.
Jünger et al (JNS)	216	Systemic Tx 119 (55.1 %): Ch 70 (32.4 %), TT 17 (7.9 %), IT 30 (13.9 %), combination 2 (0.9 %)	RT 199 (92.1 %): WBRT 64 (29.6 %), fRT 103 (47.7 %), SRS 32 (14.8 %), combination 32 (14.8 %)	Median 12.7 mos (0–88)	GTR: 199 (92.1 %) STR: 16 (7.4 %) Unclear: 1 (0.4 %)	NA	OS: **No**, GTR; KM log-rank test p = 0.343 No median OS GTR vs STR reported. PFS: NO, GTR; KM analysis, log-rank = 0.588	GTR complete cytoreduction – no influence on LC or OS
McHugh et al	110	BRAF inhibitors PD1 inhibitor	WBRT SRS	Median OS 8.1 mos (95 % CI 6.9 to 9.4)	GTR: 88 (80.0 %) STR: 19 (17.2 %) Biopsy: 3 (2.8 %)	NA	OS: **Yes**, GTR; KM log-rank p = 0.032. GTR vs STR: 9.1 (95 % CI, 7.7 to 10.5) vs 4.7 mos (95 % CI, 3.2 to 6.3)	Only KM a log-rank Mantel-Cox analysis. GTR, BRAF + status, and TT and/or IT favourable prognostic factors for OS.
Eitz et al	558	NA	No	21.2 mos (95 % CI, 18.1–24.2)	GTR: 455 (78.3 %) STR: 126 (21.7 %)	NA	OS: **Yes*** (only UV), GTR; MV Cox regression No 1.19 (0.89–1.59), p = 0.03, UV Cox regression: 1.43 (1.11–1.85), p = 0.007 LR: No, GTR; UV Cox regression: HR 0.66 (95 %CI 0.42–1.05)	OS: MV Cox regression: KPS >=80 %, 22 to 33 days btw resection and RT, controlled primary tumor associated w/ prolonged OS. LC: MV Cox regression: single BM, controlled primary tumor.
Schödel et al	805	Systemic Tx: 398 (53.1 %) Ch: 42.9 %; Molecular Tx 57.1 %	RT 613 (81.7 %)	10.9 mos (95 % CI 9.5–12.5).	GTR: 643 (79.9 %) STR: 162 (20.1 %)	NA	OS: **No,** GTR**;** KM analysis log-rank p = 0.184. No HR, no median OS reported.	Post-op clinical status, post-op RT, systemic Tx, controlled systemic disease, < 4 BM independent predictors for prolonged OS. EOR ↑ OS only if singular or solitary BM.
Olesrud et al	64	NA	IORT: 64 (100 %)	Mean OS 10.4 mos (95 % CI, 8.5 to 12.2)	NA	No	OS: **Yes**, RV; KM analysis, log-rank p = 0.05. Mean OS for measurable RV 5.6 mos (95 % CI, 3.7 to 7.4) vs 9.5 mos (95 % CI, 6.2 to 12.7) for non-measurable RV vs 12.0	KM log-rank analysis, no RV on post-op MRI independent favorable prognostic factor for OS.

(*continued on next page*)

Table 2 (*continued*)

Author	N. Patients	Systemic treatment	Local treatment	OS/PFS	GTR/STR/ EOR/RV	Total IC TB	EOR/RV association with OS/PFS?	Notes
							mos (95 % CI, 9.4 to 14.5) for no RV.	
Calluaud et al	120	Ch: 43 (35.8 %) nos	WBRT: 62 (51.7 %) SRS: 36 (30.0 %)	Mean (min–max) OS 7.9 mos (1.0–123.2)	GTR: 85 (70.8 %) STR: 35 (29.2 %)	NA	OS: **Yes**, GTR; MV Cox analysis, $p < 0.001$, HR 0.45 (95 % CI, 0.28 to 0.72). UV, $p < 0.001$ (unclear statistics) Mean OS GTR vs STR: 12.0 vs 5.6 mos	MV Cox regression, GTR, edema/tumor ratio independent favorable prognostic factors for OS; poor WHO status, low GPA, no post-op RT unfavorable prognostic factors.
She et al	122	Ch: 54 (44.3 %) nos	WBRT: 55 (45.8 %)	Median OS 9.8 mos (95 % CI, 9.1 to 10.6)	GTR: 49 (35.2 %) STR: 73 (59.8 %)	NA	OS: **Yes**, GTR; KM analysis, log-rank $p < 0.001$. Median OS 12.6 months (95 % CI, 11.4 to 13.7) for GTR vs 8.7 (95 % CI, 7.8 to 9.6) for STR	MV Cox regression: high KPS, RPA class I and II, post-op Ch independent favorable prognostic factors for OS.
Sivasanker et al	124	NA	WBRT: 64 (51.6 %)	Median OS 11.2 mos Median PFS 8.9 mos	GTR: 118 (95.2 %), NTR: 4 (3.2 %), STR; 2 (1.6 %)	NA	OS: **No**, EOR; Wilkoxon sum rank test, $p = 0.39$. PFS: **No**, EOR; Wilkoxon sum rank test, $p = 0.19$.	KM log-rank analysis, MV Cox regression: WBRT only favorable prognostic factor for OS
Antuna et al	71	NA	77.5 %	Mean OS 20.38 mos	GTR 90.1 % STR 9.9 %	Yes	OS: **Yes**, GTR; KM analysis, log-rank $p = 0.001$ No median OS for GTR vs STR reported.	Increased OS at 6 mos ($p < 0.05$) and OR of 2.9 at 12 mos. Incomplete reporting.
Kerschbaumer et al	143	NA	WBRT: 106 (74.1 %) SRS: 14 (9.8 %)	Median OS (min–max) 14.9 mos (1–116)	GTR: 101 (70.6 %), STR: 26 (18.8 %), Biopsy: 16 (11.2 %)	NA	OS: **No**, GTR; KM analysis, log-rank $p = 0.24$; EOR associated w/ increased OS. GTR/STR vs biopsy increased OS.	Surgical resection, pre-op RPA class, post-op WBRT, female sex favorable prognostic factors for OS in the UV analysis; unclear results about MV analysis reported.
Kamp et al	36	NA	WBRT (pre-op): 56.8 %, local irradiation: 21.6 %, both2.8 %	Median OS 12.9 ± 2.3 mos	GTR 15/37 (40.5 %) STR 14/37 (37.8 %) Unclear 8/37 (21.6 %)	NA	OS: **No,** GTR vs STR vs Unclear: KM analysis, log-rank $p > 0.05$. No median OS reported LR: **No**, GTR vs STR vs Unclear: KM analysis, log-rank, $p = 0.36$, Cox regression analysis $p = 0.34$	Only recurrent BM. No effect of EOR on OS. LR, effect only at Pearson Chi-square test, no effect at log-rank analysis or Cox regression.
Rava et al	87	Ch: 63 (72.4 %) nos	SRS: 81 (93.1 %) WBRT + SRS: 6 (6.9 %)	Median OS 14.3 mos	GTR: 54 (62.1 %) STR: 33 (37.9 %)	NA	OS: **No**, GTR; KM analysis log-rank $p = 0.34$ PFS: **No**, GTR; KM analysis log-rank $p = 0.27$	UV analysis: only presence of EC disease, CNS failure prognostic factors for OS. MV analysis not reported.
Gazzeri et al	57	2 Ch nos	28 RT nos	Median OS 9.0 mos (95 % CI, 4.0 to 14.0)	Single BM: 24 (NA) GTR: 14, STR: 10 Multiple BM: 19 (NA) All: 14. Symptomatic: 5	No	OS: **Yes**, EOR; KM analysis, log-rank $p < 0.05$. Mean OS GTR vs STR: 21.2 (95 % CI, 15.7 to 26.8) vs 10.1 mos (95 % CI, 5.1 to 15.1) Resection all lesions vs symptomatic ones, $p = 0.028$: 17.8 (95 % CI 12.0–23.6) vs 10.1 mos (95 % CI 5.1–15.1).	MV Cox regression: multiple BM, RPA class 2, RPA class 3 associated w/ ↓OS. Inconsistent reporting
Lee et al	157	NA	WBRT: 109 (69.4 %) SRS: 17 (10.8 %)	Median OS 19.3 mos	GTR: 119 (75.8 %) STR: 38 (24.2 %)	NA	OS: **Yes**, STR: KM analysis, log-rank $p = 0.016$; MV Cox regression analysis $p = 0.029$, HR 1.68 (95 % CI, 1.05 to 2.66). Median OS GTR vs STR: 20.4 vs 15.1 mos.	GTR, stable primary EC disease, lower RPA class favorable prognostic factors for OS. No others predictive incl. histology, type of primary cancer, and multiplicity.
Schackert, Schmiedel et al	67	NA	WBRT after 1st OP: 25 (37.3 %) WBRT only after 2nd OP: 22 (32.8 %)	Median OS 7.5 mos (95 % CI, 6.2–8.8)	GTR: 39 (58.2 %) STR: 28 (41.8 %)	Yes	OS: **Yes,** GTR: KM analysis, log-rank $p = 0.008$ 39 (58.2) 11.1. MV not significant, not reported. Median OS GTR vs STR: 11.1 (95 %CI 5.0–17.2) vs 4.8 (95 %CI 0.8–8.7) mos	Only recurrent. Single lesion, GTR, and time to recurrence (significantly influenced by WBRT after first Op) prognostic for survival.

(*continued on next page*)

Table 2 (*continued*)

Author	N. Patients	Systemic treatment	Local treatment	OS/PFS	GTR/STR/ EOR/RV	Total IC TB	EOR/RV association with OS/PFS?	Notes
Schackert, Lindner et al	127	NA	Pre-op WBRT: 10 (8.9 %), Post-op WBRT: 96(75.6 %)	Median OS: 6 mos	GTR: 29 (22.8 %) STR: 98 (77.2 %)	Yes	OS: **Yes*,** complete cytoreduction: Only in the first year after operation p = 0.039. Median OS complete vs incomplete cytoreduction: 10.6 (95 %CI 4.0.-17.2) mos vs 5.8 (95 %ci 4.1–7.6)	Favorable factors for ↑ OS were complete resection of all lesions, <= 4 BM, RPA-class, adjuvant WBRT.
Hong et al	51	Ch: 39 (76.5 %) nos	Post-op RT: 17 (33.3 %) nos	Median OS 10.8 mos.	GTR: 13 (25.5 %) STR: 38 (74.5 %) GTR vs STR: 10.8 vs 6.5 mos.	Yes	OS: **No,** GTR; MV Cox analysis p = 0.296, HR: 1.52 (95 %CI 0.69–3.32): p = 0.97. UV Cox analysis P = 0.968, HR 1.02 (95 %CI 0.50–2.07) Median OS 6.5 months vs. 10.8 months, p = 0.97	No difference in OS btw resection of all lesions vs only symptomatic ones.
Schödel et al	206	post-op Ch: 35.4 %	WBRT: 64.6 %	Median OS 6.3 mos	92.2 % GTR 7.8 % STR	NA	OS: **No,** GTR; MV Cox regression analysis, p = 0.114, HR 0.75(95 % CI 0.3–1.8). UV NA	MV analysis: poor RPA, short time interval btw initial diagnosis of first BM independent factors for decreased OS.
Auslands et al	36	NA	WBRT: 7 (19.4 %)	Mean OS 7.29 mos (95 % CI, 4.60 to 9.98)	GTR: 15 (41.7 %) STR: 21 (58.3 %)	Yes	OS: **Yes***, GTR; Mann Whitney-*U* test, p = 0.018; no association in the MV Cox analysis.	MV Cox regression: only lower RPA class favorable prognostic factor for OS.
Limbrick et al	15	NA	SRS only: 9 (60.0 %) SRS + WBRT: 6 (40.0 %)	Median (min–max) OS 20.0 mos (5.0–68.0)	GTR: 12 (80.0 %) STR: 5 (32.3 %)	Yes	OS: **No**, GTR: log-rank p = 0.44. Median OS GTR: 20 (5–68) mos.	Resection + SRS. No OS benefit in GTR.
Yoo et al	94	NA	Post-op RT: 29 (30.9 %) No post-op RT: 65 (69.1 %)	11.0 months (95 % CI 8.1–13.9 months).	GTR: 51 (54.2 %) MTR: 43 (45.8 %)	Yes	OS: **No,** MTR: UV and MV Cox regression analysis OS GTR vs MTR, p = 0.974, HR 1.01 (0.57–1.78). Median OS MTR vs GTR: 10.3 vs 11.0 mos. LR: MTR vs GTR: **Yes**, MV Cox analysis, p = 0.003 HR 3.14 (95 % CI 1.47–6.72)	Only RPA associated w/ ↑ OS at MV analysis. 2-yr OS rate ↑ in MTR group, LR rate ↓ in MTR group at MV analysis. 1-yr OS rate: 49.2 % vs 42.8 % p = 0.54. 2-yr OS: 27.3 % vs 3.8 %, p = 0.001. 1-year LR rate: 29.1 % vs 58.6 %. 2-yr LR rate: 29.1 % vs 63.2 %, p = 0.01.
Rades et al	201	NA	WBRT: 99 (49.3 %) WBRT + boost: 102 (50.7 %)	Median OS 14.5 mos	GTR: 131 (65.2 %) STR: 70 (34.8 %)	NA	OS: **Yes,** GTR vs STR; MV Cox regression, p = 0.001, HR 1.80 (95 % CI, 1.26 to 2.55). UV NA. LR: **Yes**; MV Cox regression, p=<0.001, HR 3.79 (95 %CI 2.34–6.22)	Only patients w/ RPA class I and II included. GTR, WBRT + boost, and interval diagnosis-WBRT favorable prognostic factors for OS. Tx regimen and GTR prognostic for LR.
Tendulkar et al	271	NA	WBRT: 233 (86.0 %) SRS: 56 (14.0 %)	Median OS 10.2 mos	GTR: 216 (79.7 %) STR: 55 (20.3 %)	Yes	OS: **No,** GTR**;** KM analysis log-rank**,** p = 0.07 GTR vs STR: 10.8 vs 7 mos.	MV Cox regression: age < 65 yrs, lack of EC BM, control of primary tumor, histology (NSCLC), SRS associated w/ ↑ OS
Paek et al	208	Pre-op Ch: 108 (51.9 %). Post-op Ch: 104 (50.0 %).	Pre-op RT: 45 (21.6 %), Post-op WBRT/SRS: 146 (70.2 %), Pre-op RS: 3(1.4 %)	Median OS 8 mos	GTR: 187 (89.9 %) STR: 2 (10.1 %)	NA	OS: **No** GTR: KM analysis, log-rank p = ns. Median OS GTR vs STR and p value NA.	MV Cox regression: KPS, RPA predictive of ↑ OS
Stark et al	177	0 (0 %)	Post-op RT: 123 (69.5 %) nos	Median OS 9.3 mos	GTR: 133 (75.1 %) STR: 41 (23.2 %) Biopsy: 3 (1.7 %)	Yes	OS: **Yes,** complete cytoreduction**;** KM analysis, log-rank, p = 0.035. Median OS GTR vs STR NA.	Parameters significantly associated w/ ↑ OS: Pre-op: age < 70 yrs, 1–3 BM. Tx-related: GTR, post-op GOS 4–5, recraniotomy.
Bindal et al	82	Ch: 7 (8.5 %) nos	WBRT: 17 (20.7 %)	Median OS 10 mos.	GTR: 52 (63.4 %) STR: 30 (36.6 %)	Yes	OS: **Yes,** complete cytoreduction: all vs. not all BM removed: KM analysis, log-rank, p = 0.003 for multiple BM and p = 0.012	OS pts w/ multiple BM (all resected) ↑ vs pts w/ multiple BM (only 1 is resected), log rank: p < 0.003, and similar

(*continued on next page*)

Table 2 (*continued*)

Author	N. Patients	Systemic treatment	Local treatment	OS/PFS	GTR/STR/ EOR/RV	Total IC TB	EOR/RV association with OS/PFS?	Notes
							for not_all_in_multipleBM vs. all_in_singleBM. No resection all BM vs resection all BM vs resection single BM 6 vs 14 vs 14 mos.	to those of single BM resected. No difference in BM recurrence btw resection all lesion and resection single BM.

Abbreviations. BM: brain metastasis; Ch: chemotherapy, CRC: colorectal cancer; EOR: extent of resection; GI: gastrointestinal; GTR: gross total resection; HR: hazard ratio; HSRT: hypofractionated stereotactic radiotherapy; ICI: immune checkpoint inhibitor; IT, immunotherapy; KM: Kaplan-Meier; MTR: microscopic total resection; NOS: not otherwise defined; NSCLC: non-small cell lung cancer; LBRT: local brain radiotherapy; PBRT: partial brain radiotherapy; RPA: recursive partitioning analysis; RV: residual volume; RT: radiotherapy; RPA: recursive partition analysis; RS: retrospective study; SRS: stereotactic radiosurgery; STR: subtotal resection; TT: targeted therapy, WBRT: whole – brain radiotherapy.
*Studies with association at univariate Cox regression not confirmed at multivariable Cox regression or with association found with statistical methods different from Kaplan-Meier analysis/log-rank test.

and 16[47–62] additional studies were included. A total of 39 studies were included in the final analysis (see Table 1)[8,9,26–62].

Study characteristics

All included studies were retrospective[8,9,26–62]. Three studies were multicentric[31,50,51] while the others were conducted at a single institution (Fig. 1A, Table 1)[8,9,26–30,32–49,52–62]. The study population ranged from 15 to 750 patients (mean = 173.5; median = 124).

Inclusion timeframes

Inclusion timeframes of included studies had start years ranging from 1984 to 2015 and end years from 1992 to 2022. Mean inclusion time was 10.9 year (median = 10; range = 3–21). A synoptic overview of the inclusion timeframe for each study is shown in Fig. 1B. Only 2 studies did not report inclusion timeframes[31,57].

Study quality/grading

After GRADE assessment of the included studies, 17 studies ranked as "moderate"[8,29,31–35,39,41,44–47,49,51,60,61], 14 ranked as "low"[9,26,27,30,36–38,40,48,50,52,54,57,59] and the remaining 8 as "very low" [28,42,43,53,55,56,58,62] (Table 1).

Primary tumors

Thirty-seven studies (94.9 %) reported data on histology of the primary tumor[8,26,28–62]. The majority (n = 28/37, 75.7 %) included mixed populations with heterogenous primary tumors (see Fig. 1C and Table 1)[26,29–32,34,35,37–44,46,47,50,51,53–55,57–62]. Nine of 37 studies (24.3 %) included only BM from a single primary tumor [8,28,33,36,45,48,49,52,56], 6 of which assessed patients with lung cancer BM[36,45,48,49,52,56], 2 melanoma BM[8,33], and 1 BM from colorectal cancer[28]. Only 2 studies did not report primary tumor histology[9,27] (Fig. 1C, Table 1 and Fig. 2 for detailed numbers of primary tumor entities across studies and within each study). Overall, 46 % (n = 2757) of the patients across all studies had BMs originating from lung cancer, followed by melanoma (14 %, n = 846, Fig. 2A,B). Heterogeneity regarding primary tumor entities was high in most of the included studies (Fig. 2C,D).

Single vs. Multiple BMs

Thirty-seven of 39 studies reported the number of BM within each patient[8,9,26–43,45–59,61,62]. The majority of studies included patients with both single and multiple BM (n = 35, Table 1, Fig. 1D) [8,9,26–28,30,31,33–57,60–62], with 3 studies including only patients with single BM[29,58,59]. With respect to surgical treatment of one or more BM, 4 studies investigated surgical resection of only one lesion [29,32,58,59]. One study reported only combined resection of multiple BMs[42]. The majority (n = 25) of papers included patients who received resection of 1 or more lesions, with heterogenous reporting thereof[8,9,26,28,31,33–37,40,43–45,48,49,52–57,60–62] while 9 of them did not clearly specified number of metastases resected in the same surgery[27,30,38,39,41,46,47,50,51].

Volumetric analysis reporting and assessment of resection

Twenty-five studies reported data of preoperative[8,9,27–32, 35–39,41–43,45–48,52,55,58–60], and 9 on postoperative volume [8,9,27,28,34,40,45–47], 8 reported both[8,9,27,28,40,45–47] (Fig. 1D, Table 2).

With respect to assessment of resection, 33 studies stratified EOR as GTR versus STR[26,27,29–57,59,60], while data on RV and EOR were reported in 8[8,9,26–28,45–47] and 2 studies[40,47], respectively. One study assessed microscopic-total resection versus GTR[58] and 2 studies assessed the resection of all versus selected lesions in patients with multiple BMs[61,62]. Information on postoperative intracranial tumor burden was heterogeneously reported, with this information at least in part deducible in 16 studies[8,9,27–29,32,42,43,45,46,52,54–56, 61,62].

Neurological performance and complications reporting

Sixteen studies reported only preoperative neurological status using either ECOG or Karnofsky performance status (KPS; see Table 2) [30,34,36,39,40,42–45,47,48,53,56,58,59,62], 1 study reported only postoperative functional status[50], while 16 studies reporting both [9,27–29,32,37,40,41,46,49,51,52,54,55,57,60]. Four studies did not report any functional performance data[26,33,35,38]. Twenty-three studies reported data on postoperative complications[26,28,29,31,32, 34–37,40–42,44,47,49,51–54,57,60–62].

Information on extracranial disease status and metastases, systemic and local treatments

Information on extracranial disease status and extracranial metastases were reported in 22[9,27,29–33,35,39,41,42,45,47,49–51, 55–57,59,60,62] and 19 studies[8,29,35–39,44,45,47–50,52–55,59 ,61], respectively (Fig. 1D, Table 2). Twenty-one studies reported data on systemic therapies[8,9,27,28,30,33,35–37,40,45,46,48,49,51, 55–57,60–62] (Fig. 1D), with only 6 of them reporting immunotherapy treatment[8,27,33,45,46,49] and 5 targeted therapies[30,32,33,45,48], with most studies generally reporting chemotherapy treatment. All but 1 [47] studies reported additional local treatment including RT – mostly either whole brain, local RT, or SRS [8,9,26–46,48–62] (Fig. 1D). Only 2 studies reported on the use of steroids[8,47] (Fig. 1D).

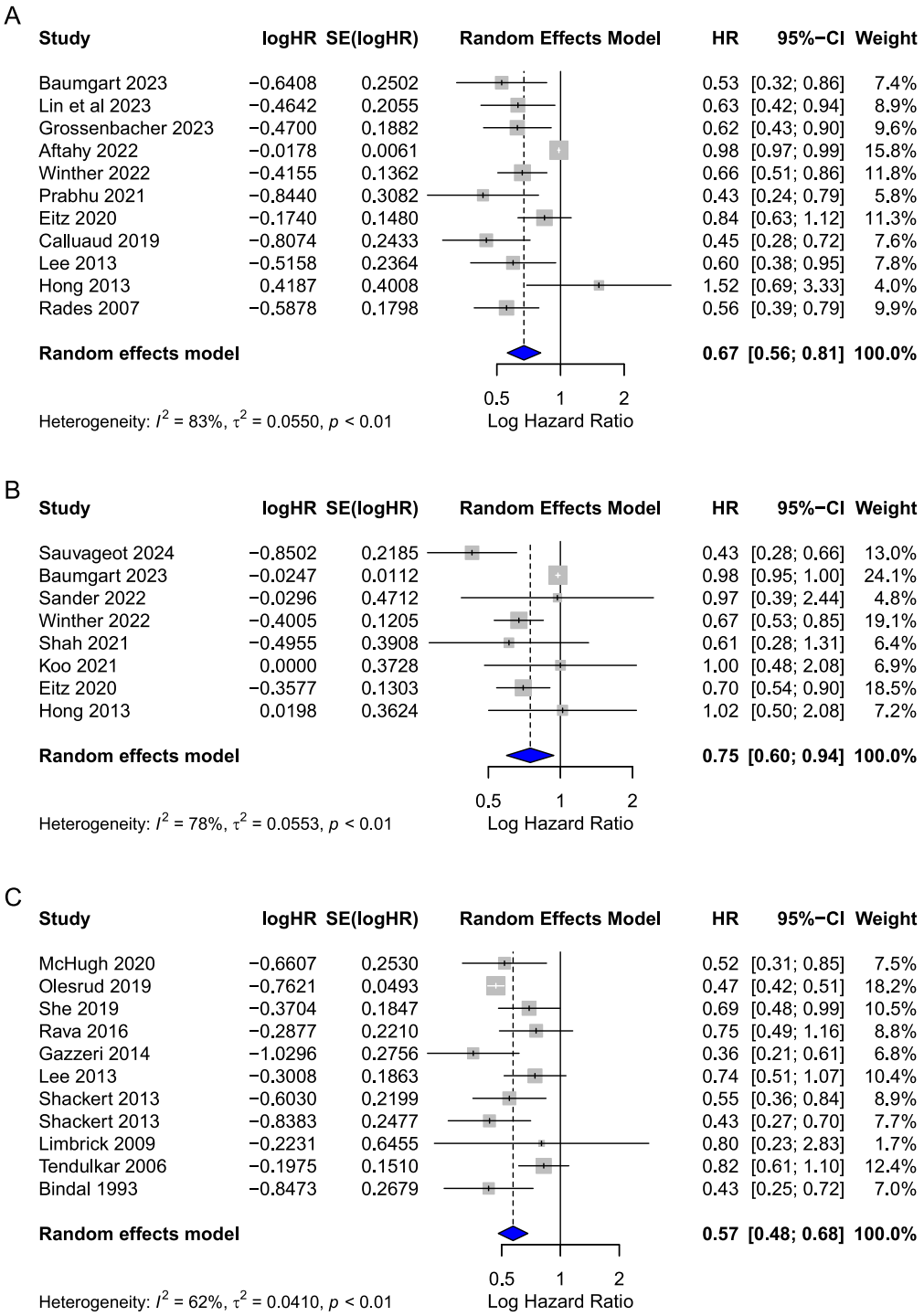

Fig. 3. Forrest plots for overall survival. Studies assessing the relationship of gross total resection with overall survival are analyzed with three different *meta*-analysis according to statistical power. A. Multivariate Cox-regression; B. Univariate Cox-regression; C. Kaplan-Meier log-rank statistics. The forest plots display the effect sizes (log HR) and 95% confidence intervals (CIs) for each study included in the individual *meta*-analysis, sorted by author. Each horizontal line represents an individual study's effect size and CI, with the study's weight indicated by the size of the marker. The blue diamond at the bottom represents the overall summary estimate and 95% CI, calculated using a random-effects model. These plots provide a visual overview of the variability in effect sizes across studies and highlight the overall trend for the log hazard ratio across the dataset, according to statistical analysis performed. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Outcome reporting

All studies reported OS as an outcome[8,9,26–62], 12 studies reported also local PFS[30–32,37,39,44,45,49,50,53,58,62].

Association of EOR and survival

Twenty-one studies reported an association of greater EOR with improved OS[9,10,26,27,29,31,33–36,40,41,44,46,47,50,52,54,55,61,62], 4 reported conflicting results[42,45,51,58] while 14 studies found no difference[28,30,32,37–39,43,48,49,53,56,57,59,60] (Fig. 1D,

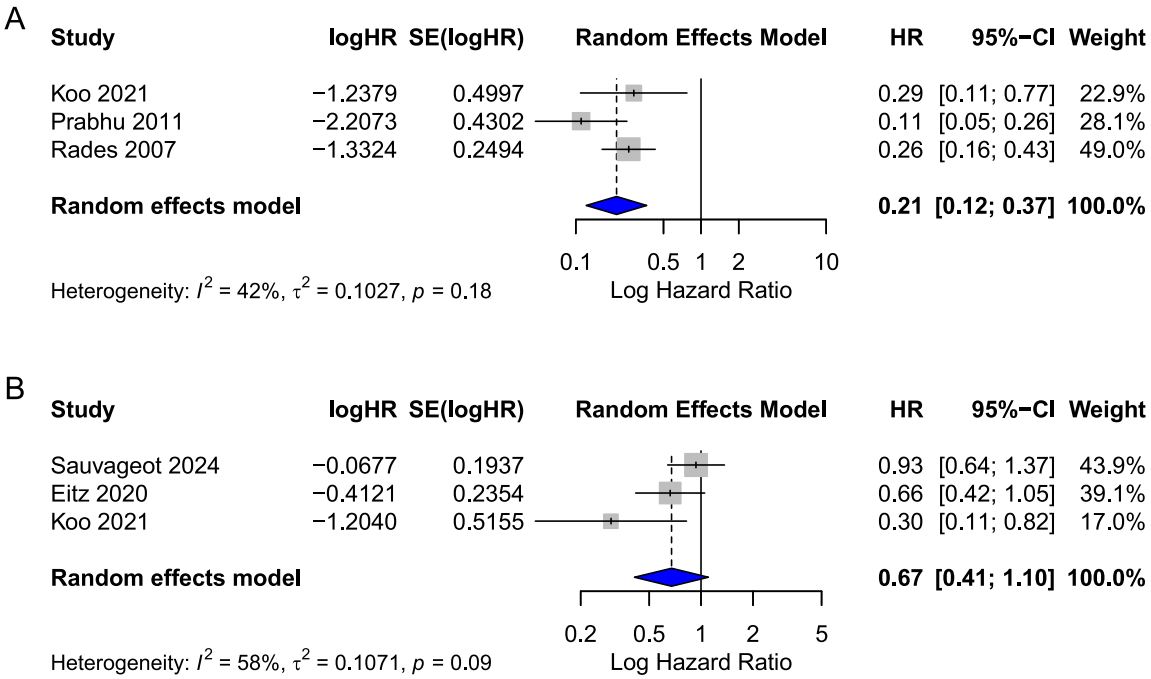

Fig. 4. Forrest plots for progression-free survival. Studies assessing the relationship of gross total resection with progression-free survival are analyzed with two different *meta*-analysis according to statistical power (A. Multivariate Cox-regression; B. Univariate Cox-regression). The forest plots display the effect sizes (log HR) and 95% confidence intervals (CIs) for each study included in the individual *meta*-analysis, sorted by author. Each horizontal line represents an individual study's effect size and CI, with the study's weight indicated by the size of the marker. The blue diamond at the bottom represents the overall summary estimate and 95% CI, calculated using a random-effects model. These plots provide a visual overview of the variability in effect sizes across studies and highlight the overall trend for the log hazard ratio across the dataset, according to statistical analysis performed. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2).

Of the studies reporting PFS, 5 found an association of more extensive resection with improved PFS[30,31,44,45,58], while 6 did not find any difference[32,39,49,50,53,62].

Association between EOR and OS/PFS – evaluation of statistical analyses

Supplementary file 1 summarizes the statistical analyses used in the different studies to assess the association of EOR and OS and PFS.

OS – random-effects meta-analysis

Random-effects *meta*-analyses were performed for studies where HRs were reported or could be estimated. Three separate *meta*-analyses were run according to the level of statistical strength (see Methods). Eleven studies reporting results of multivariate Cox-regression models [9,27,29,31,35,41,44,46,47,50,56] were pooled in the first of the three *meta*-analyses. Covariates used to build the models were heterogenous among included studies and most commonly included sex[29,31,46], age[9,27,29,31,46,47] neurological status[27,29,44,46,56], preoperative tumor burden[9,27,46], information on primary tumor control or extracranial status[29,31,47,56] and RT[35,44]. Molecular classification, histology, steroid treatment, and anatomic locations of BMs were rarely reported in multivariate Cox-regression models. With respect to univariate Cox-regression models, 8 studies reported HRs [28–30,45,46,48,50,56] and were pooled for the second *meta*-analysis. Fourteen studies reported log-rank test results[33,34,36,39–41,43,54–56,59–62]. For eleven of them, HRs could be derived as described in the methods.

In the first *meta*-analysis which included studies reporting multivariate analysis, the pooled HR of GTR vs. STR was 0.67 (95 %CI 0.56–0.81, Fig. 3A), indicating a statistically significant association of GTR with longer OS ($p < 0.001$). Heterogeneity analysis revealed a Q statistic of 59.04 ($p < 0.001$) and an I2 value of 83 %, suggesting substantial heterogeneity among the studies. Egger's test resulted in a p-value < 0.0001, indicating evidence of publication bias.

In the second *meta*-analysis which included studies univariate analysis, the pooled HR of GTR versus STR for OS was 0.75 (95 %CI 0.60–0.94), indicating a statistically significant association of GTR with better OS ($p = 0.019$, Fig. 3B). Heterogeneity analysis revealed a Q statistic of 31.47 ($p < 0.001$) and an I2 value of 78 %, suggesting substantial heterogeneity among the studies. Egger's test in a p-value of 0.06, indicating lack of significant publication bias.

In the third *meta*-analysis which pooled studies whose HRs were derived from log-rank tests, the pooled HR of GTR for survival was 0.57 (95 %CI 0.48–0.68, Fig. 3C), indicating a statistically significant association of GTR with better OS ($p < 0.0001$). Heterogeneity analysis revealed a Q statistic of 26 ($p < 0.01$) and an I2 value of 61.5 %, suggesting moderate heterogeneity among the studies. Egger's test resulted in a p-value of 0.17, indicating lack of significant publication bias. The results of the three *meta*-analyses are visually summarized using forest plots (Fig. 3), with corresponding funnel plots (Supplementary Fig. 2) and leave-one-out analysis plots (Supplementary Fig. 3).

Local PFS – random-effects meta-analysis

Of the 12 studies reporting an association of EOR on PFS, only the minority reported necessary data to be pooled in a *meta*-analysis and 6 of them were included for quantitative analysis. In detail, 3 studies performed multivariate Cox-regression analysis[30,31,44] and 3 studies reported univariate Cox-regression[30,45,50].

In the first *meta*-analysis which included studies reporting multivariate analysis, the pooled HR of GTR for PFS was 0.21 (95 %CI 0.12–0.37, Fig. 4A), indicating a statistically significant increase in PFS associated with GTR ($p < 0.001$). Heterogeneity analysis revealed a Q statistic of 3.43 ($p = 0.18$) and an I2 value of 42 %, suggesting low

heterogeneity among the studies. Egger's test for publication bias resulted in a p-value of 0.72, suggesting absence of publication bias. In the *meta*-analysis which pooled studies reporting univariate analysis, the pooled HR of GTR for PFS was 0.67 (95 %CI 0.41–1.10, Fig. 4B), and therefore showed no statistically significant association of PFS and GTR ($p = 0.11$). Heterogeneity analysis revealed a Q statistic of 4.73 ($p = 0.09$) and an I2 value of 58 %, suggesting moderate heterogeneity among the studies. Egger's test for publication bias resulted in a p-value of 0.19, excluding significant publication bias. Corresponding funnel plots and leave-one-out analysis plots are shown in Supplementary Figs. 4 and 5).

Subgroup meta-analyses

We confirmed the feasibility of subgroup *meta*-analyses based on recent publication timeframe and homogenous primary tumor histology. In a first subgroup analysis we included only studies published in the last 10 years leaving 17 studies[9,27–31,33–36,39,40,45–48,50] for inclusion (8 multi-, 4 univariate Cox-regression and 5 log-rank tests). In this analysis, association of GTR with longer OS was confirmed (HR 0.62, 95 %CI 0.53–0.72, $p < 0.001$) yet showing also high heterogeneity (Q statistic 305.9, I^2 value of 95 %, $p < 0.001$). Of the 6 studies including only lung cancer BM, 4[36,45,48,56] could be included (3 univariate Cox-regression, 1 log-rank test). No statistically significant association between GTR and OS was found (HR 0.68, 95 %CI 0.42–1.09, $p = 0.10$) with moderate heterogeneity of the included studies (Q statistic 8.24, I^2 value of 63.6 %). A subgroup *meta*-analysis for melanoma BM was not judged feasible ($n < 3$, insufficient outcome reporting). Other possible subgroup *meta*-analyses (see Methods) were judged unfeasible due to lack of sufficient data reporting. The subgroup *meta*-analyses forest, leave-one-out analyses and funnel plots are reported in Supplementary Fig. 6.

Discussion

Neurosurgical resection of BM is considered in selected clinical scenarios: ranging from cases of diagnostic uncertainty in patients without a known primary tumor, to therapeutic objectives in cases with single lesions and controlled extracranial disease or with large tumors causing mass effect and/or neurological symptoms[2,63].

However, compared to patients with primary brain tumors, where there is substantial evidence of the beneficial survival effects of a more extensive neurosurgical resection[4], its role in patients with BM remains debated. As the treatment modalities for BM patients are constantly changing, the present review aimed to systematically assess the existing evidence investigating the relationship of a more extensive surgical resection of BM with survival outcome.

In contrast to primary brain tumors, the available evidence consists entirely of retrospective, mostly single center studies, and no randomized controlled trials are available. Nevertheless, there has been a noticeable increase in the number of studies on this topic in the last decade, which is in line with the very dynamic field of interdisciplinary treatment of BM patients. Although our *meta*-analysis showed an overall association of GTR with OS and PFS in all studies reporting multivariate analysis, our literature review revealed substantial heterogeneity in study quality, statistical reporting, primary tumors included, and reporting of potentially important covariates.

Primary tumor entities

Almost all of the identified studies consisted of heterogeneous cohorts of patients with different primary tumors. This is surprising because it neglects not only the intrinsic differences in primary tumor biology, but also the differences in available systemic treatments and their efficacy in different tumor entities, as well as the different sensitivities to RT. Notably, several of the studies which analyzed only BM from distinct primary tumor entities did not report a survival benefit upon a more extensive resection, notably for lung tumor BM[48,49,56]. Conversely, reports from other groups focusing only on lung cancer BM found an OS benefit in patients with GTR compared with STR based on KM analysis[36,52] or based on postoperative RV in a univariate analysis which did not show statistically significance in multivariate model [45]. Similarly, one study showed no OS benefit upon a more extensive resection in patients with BM from gastrointestinal origin[28]. On the other hand, for melanoma BM our review identified two studies[8,33] which showed a OS benefit in patients undergoing a more extensive surgical BM resection in either KM survival curve analysis for GTR vs. STR and biopsy only [33] or in uni-, multivariate Cox-regression and KM analysis[8]. After completion of the literature review another study focusing on a single BM entity was published[64]: in a bi-centric retrospective cohort analysis, it did not show a prognostic effect of a more extensive surgical resection in patients with breast cancer BM. However, other characteristics such as molecular subtypes and extracranial disease status were prognostically important for OS and a subgroup of patients with controlled extracranial disease might benefit from a more extensive resection.

Overall, our analysis showed a strong predominance of lung cancer BM (almost 50 %) in all studies. This quantitative dominance of lung cancer BM was also present in study cohorts composed of different primary tumor entities, which may further bias the results of these studies considering the different inherent tumor and treatment characteristics.

Treatments beyond surgery and steroids

Another important factor that arises from the different primary tumor entities that often metastasize to the brain and that must be considered when analyzing the role of surgical tumor burden reduction is the different systemic and local adjuvant treatment regimens that patients may receive pre- or postoperatively: With the introduction of targeted and immunotherapies in the setting of central nervous system metastases of solid tumors, numerous studies have shown that e.g. immune checkpoint inhibitors can induce intracranial response in patients with BM, especially in melanoma[65–67]. Although our analysis showed 21 of 39 studies reporting any systemic treatments in their cohorts, only the minority of studies provided information on the administration of targeted therapies or immunotherapies which is not surprising since many of those agents found their way into common clinical practice only recently. However, among the studies which found a significant association between EOR or postoperative tumor volume and survival, almost none included pre- or postoperative targeted/immunotherapy in multivariate survival models to assess for potential bias. Only one study on resection of melanoma BM included the perioperative use of immune checkpoint inhibitor therapy in their analysis and indeed reported a benefit of a more extensive surgical resection in patients which were immune checkpoint inhibitor-naive and received postoperative double checkpoint inhibitor therapy (ipilimumab + nivolumab)[8]. Conversely, local RT/SRS is a current mainstay of the multimodal treatment of BM and its use has been extensively reported in all studies. On the other hand, we must acknowledge that despite the reported association of EOR with improved survival, the extent to which other local therapies and their effect on tumor control confound the present findings. It should be noted that major studies investigating the role of postoperative RT in improving outcome in BM patients included only patients who received gross total resection[68] and that, unfortunately, the inclusion of RT as a covariate in multivariate analyses of trials investigating the role of surgical tumor volume reduction remains rare[35,44].

Another important issue to consider is the use of steroids, which are often used to treat perilesional edema in symptomatic BM patients. Several studies have shown that patients with BM who receive immune-checkpoint inhibitor treatment with concomitant steroid administration

have shorter survival rates[69,70]. Therefore, the effect of more extensive neurosurgical resection in some patients with BM who are undergoing or are eligible for postoperative immunotherapy may be an epiphenomenon of reduced steroid dependency that optimizes the efficacy of immune-checkpoint inhibitor therapy. For example, one study suggests upfront resection of melanoma BM because their data showed improved survival in immune-checkpoint inhibitor-naive patients who underwent surgical resection prior to immune-checkpoint inhibitor therapy. This may be due to faster steroid tapering, although the actual steroid use was not reported[71]. Of the analyzed studies, only 2 reported the use of steroids either perioperatively (highest dose of dexamethasone equivalent within 4 weeks of BM resection[47]) or postoperatively (steroid intake 4 weeks after BM resection[8]). Interestingly, both studies showed that the extent of surgical BM resection was independent from steroid intake/dose associated with OS in a multivariate Cox-regression model.

What to measure – extent of resection or residual volume?

When evaluating the role of neurosurgical resection in relation to cytoreduction in patients with brain tumor patients, comparisons between studies are further complicated by the metric used to assess the extent of surgical resection: while many authors have relied on the EOR with either volumetric analysis of the resulting preoperative and postoperative contrast-enhancing tumor or categorical cutoffs – usually GTR and STR – reflecting no residual or residual contrast-enhancing tumor mass on postoperative MRI, they usually refer to the targeted lesions that were surgically treated. From a surgical and oncological point of view, this may make sense in patients with BM with solitary lesions, but since BMs are often multifocal, and most of the identified articles also included patients with multiple BMs, the more appropriate measure seems to be the EOR relative to the total intracranial tumor burden, or even better, the remaining intracranial tumor volume as a measure of metastatic tumor burden in the brain. The concept of total intracranial tumor burden seems to be receiving increasing attention, and the more recent studies included in our review have increasingly compared the EOR to the total intracranial tumor burden[8,9,28,45,46].

In addition to reduced steroid dependency in BM patients with reduced total intracranial tumor volume, recent studies suggest that the efficacy of immune-checkpoint inhibitor also depends on the total body tumor burden, e.g. assessed by PET-CT, and that increased tumor burden is associated with a worse immune-checkpoint inhibitor response[72]. Whether this also applies to intracranial tumor burden and immune-checkpoint inhibitor efficacy certainly needs further investigation.

Resection of multiple lesions

If surgical tumor load reduction in BM patients results in a survival benefit, as suggested by the studies analyzed in the present review with all the limitations mentioned above, the question arises whether multiple resections should be performed in patients with multifocal BM dissemination. Although resection of multiple BM within the same operation, even through multiple craniotomies, is rarely performed, there are numerous reports, mainly focused on adverse event rates, surgical and functional outcomes, which nevertheless indicate that this can be performed safely and that adverse event rates are not significantly increased compared to patients who undergo resection of only one BM[73–77]. However, the role of resection of multiple BMs in the same patient in terms of oncologic outcomes has not been adequately investigated and needs to be addressed in future studies in the context of the primary disease, but also in the context of other local and tumor-specific treatment modalities.

Resection of recurrent BM

The situation of recurrent BM after initial RT or previous resection represents a special situation that patients and clinicians are increasingly confronted with. In our review only very few studies included only patients with recurrent lesions making it difficult to draw conclusions from such limited data. Particularly in patients who initially received RT or immune-checkpoint inhibitor, distinguishing between adverse treatment effects and intracranial disease progression is often a major challenge[78].

Apart from survival outcome: other indications for neurosurgical BM resections

Beyond the role of surgical tumor burden reduction in BM patients studied and highlighted here, neurosurgical BM resection may indirectly improve survival outcomes in patients with symptomatic lesions, not only by improving symptoms and functional performance status, but also by allowing them to receive postoperative systemic treatments that they would not otherwise qualify for[51].

In addition, there are many unanswered questions about the ideal treatment algorithms for BM patients, including the timing of surgical resection and its interaction with other local and systemic treatments. For example, SRS is usually performed after surgical resection to reduce local recurrence rates[2,63], but there is a rationale for upfront SRS before surgical resection, e.g. to shorten time to next treatment, reduce leptomeningeal spread of tumor cells, decrease wound complications, and allow for faster steroid tapering[79–82]. Furthermore, surgical BM resection may also provide a window of opportunity for targeted systemic treatments, as it has been investigated that BM may harbor various driver mutations that are not present in extracranial metastases or the primary tumors[83]; however, the overall frequency of altered targetable mutations in BM and the clinical impact is still unclear and needs to be addressed by future studies.

Limitations

Despite a thorough review of the available evidence, this study has limitations. First, the available literature has a low overall level of evidence, as it is almost entirely composed of retrospective monocentric studies. In addition, most studies include heterogeneous BM cohorts of different primary tumors, which represents a significant confounding factor given their different inherent behavior and treatment responses. Instead, the management of BMs is mainly tailored to the individual patient, taking into account various individual factors such as histopathological features, systemic tumor burden, availability of targeted/immuno-therapies, functional status, and local protocols. This increases the granularity and heterogeneity of the data, making direct and reliable pooled analysis complex and challenging. Publication and reporting bias may have influenced the selective reporting of positive results, biasing the results of the *meta*-analysis performed towards a positive effect of GTR over STR. Nevertheless, the results of negative studies that could not be pooled in the *meta*-analyses were comprehensively reported to provide the most accurate and unbiased representation of the available evidence.

Conclusions

Most studies identified in our systematic literature review suggest that a more extensive neurosurgical resection in BM patients is associated with improved survival, but this effect may be tumor-specific and dependent on various other confounding factors that may reflect systemic disease status or interact with subsequent systemic treatment options. Generalized conclusions for all patients with BM are likely to be oversimplified, as they do not consider the diverse disease characteristics and multidisciplinary treatment landscape of the various primary tumors.

As in primary brain tumors, future research efforts should focus on a more primary disease-specific and even molecular subgroup focused

Table 3
Methodological considerations and analysis suggestions for future studies assessing role of surgical resection in patients with brain metastases.

General
• Focus on BM from distinct primary tumor entities (with molecular characterization)
Covariates to report
• Primary tumor entities
• Single/multiple BMs
• First/recurrent surgery for BM
• Status of extracranial disease
• ICI therapy, other targeted/systemic therapies
• Steroid intake postoperative/FU
• RT modality postoperative
• Residual tumor volume/EOR/GTR vs. STR (relative to total intracranial tumor volume)
Statistical reporting
• Uni-/Multivariable Cox-proportional hazard model: HR, 95 %CI, p-values
• Kaplan-Meier/Log-rank statistics: median OS/PFS, p-values
• For subgroup analysis median OS of subgroups

investigation of clinical cohorts regarding the survival effect of neurosurgical resection extensiveness. In addition, relevant confounding factors should be taken into account in study design, descriptive and statistical reporting, while always including negative study results. With this objective, we propose a set of minimal requirements that should be included in future studies to address and effectively counteract the discussed potential sources of bias (Table 3).

We envision that the extent of BM resection, especially in patients with multiple lesions and beyond "classical" indications such as symptomatic mass effect, will be individualized and tailored to the primary tumor features (including molecular analysis), patient's status and systemic disease stage, as well as available adjuvant treatment characteristics.

Funding

No specific funding was received for this study. Vittorio Stumpo is supported by the UZH Candoc Grant 2023 (FK-23–048).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ctrv.2025.102981.

References

[1] Achrol AS, Rennert RC, Anders C, Soffietti R, Ahluwalia MS, Nayak L, et al. Brain metastases. Nat. Rev. Dis. Primers 2019;5:1–26. https://doi.org/10.1038/s41572-018-0055-y.
[2] Rhun EL, Guckenberger M, Smits M, Dummer R, Bachelot T, Sahm F, et al. EANO–ESMO Clinical Practice guidelines for diagnosis, treatment and follow-up of patients with brain metastasis from solid tumours☆. Ann. Oncol. 2021;32:1332–47. https://doi.org/10.1016/j.annonc.2021.07.016.
[3] Aizer AA, Lamba N, Ahluwalia MS, Aldape K, Boire A, Brastianos PK, et al. Brain metastases: A Society for Neuro-Oncology (SNO) consensus review on current management and future directions. Neuro-Oncology 2022:noac118. https://doi.org/10.1093/neuonc/noac118.
[4] Karschnia P, Gerritsen JKW, Teske N, Cahill DP, Jakola AS, van den Bent M, et al. The oncological role of resection in newly diagnosed diffuse adult-type glioma defined by the WHO 2021 classification: a Review by the RANO resect group. Lancet Oncol. 2024;25:e404. https://doi.org/10.1016/S1470-2045(24)00130-X.
[5] Stupp R, Mason WP, van den Bent MJ, Weller M, Fisher B, Taphoorn MJB, et al. Radiotherapy plus Concomitant and Adjuvant Temozolomide for Glioblastoma. N. Engl. J. Med. 2005;352:987–96. https://doi.org/10.1056/NEJMoa043330.
[6] Weller M, van den Bent M, Preusser M, Le Rhun E, Tonn JC, Minniti G, et al. EANO guidelines on the diagnosis and treatment of diffuse gliomas of adulthood. Nat. Rev. Clin. Oncol. 2020:1–17. https://doi.org/10.1038/s41571-020-00447-z.
[7] Suh JH, Kotecha R, Chao ST, Ahluwalia MS, Sahgal A, Chang EL. Current approaches to the management of brain metastases. Nat. Rev. Clin. Oncol. 2020;17:279–99. https://doi.org/10.1038/s41571-019-0320-3.
[8] Voglis S, Schaller V, Müller T, Gönel M, Winklhofer S, Mangana J, et al. Maximal surgical tumour load reduction in immune-checkpoint inhibitor naïve patients with melanoma brain metastases correlates with prolonged survival. Eur. J. Cancer 2022;175:158–68. https://doi.org/10.1016/j.ejca.2022.08.020.
[9] Lin J, Kaiser Y, Wiestler B, Bernhardt D, Combs SE, Delbridge C, et al. Cytoreduction of Residual Tumor Burden is Decisive for Prolonged Survival in patients with Recurrent Brain Metastases-Retrospective Analysis of 219 patients. Cancers (Basel) 2023;15:5067. https://doi.org/10.3390/cancers15205067.
[10] Voglis S, Padevit L, van Niftrik CHB, Kälin V, Beyersdorf B, Da Mutten R, et al. Safety of microneurosurgical interventions for superficial and deep-seated brain metastases: single-center cohort study of 637 consecutive cases. J. Neurooncol 2023. https://doi.org/10.1007/s11060-023-04478-1.
[11] Proescholdt MA, Schödel P, Doenitz C, Pukrop T, Höhne J, Schmidt NO, et al. The Management of Brain Metastases—Systematic Review of Neurosurgical Aspects. Cancers 2021;13:1616. https://doi.org/10.3390/cancers13071616.
[12] Moher D, Liberati A, Tetzlaff J, Altman DG. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. BMJ 2009;339:b2535. https://doi.org/10.1136/bmj.b2535.
[13] Guyatt GH, Oxman AD, Vist GE, Kunz R, Falck-Ytter Y, Alonso-Coello P, et al. GRADE: an emerging consensus on rating quality of evidence and strength of recommendations. BMJ 2008;336:924–6. https://doi.org/10.1136/bmj.39489.470347.AD.
[14] R Core Team. R: A language and environment for statistical computing 2024.
[15] Bland JM, Altman DG. Survival probabilities (the Kaplan-Meier method). BMJ 1998;317:1572–80. https://doi.org/10.1136/bmj.317.7172.1572.
[16] Parmar MKB, Torri V, Stewart L. Extracting summary statistics to perform meta-analyses of the published literature for survival endpoints. Stat. Med. 1998;17:2815–34. https://doi.org/10.1002/(SICI)1097-0258(19981230)17:24<2815::AID-SIM110>3.0.CO;2-8.
[17] Tierney JF, Stewart LA, Ghersi D, Burdett S, Sydes MR. Practical methods for incorporating summary time-to-event data into meta-analysis. Trials 2007;8:16. https://doi.org/10.1186/1745-6215-8-16.
[18] Altman DG, Bland JM. Interaction revisited: the difference between two estimates. BMJ 2003;326:219. https://doi.org/10.1136/bmj.326.7382.219.
[19] Higgins JP, Green S. Cochrane Handbook for Systematic Reviews of Interventions : Cochrane Book Series n.d.
[20] Deeks JJ, Altman DG, Bradburn MJ. Statistical Methods for Examining Heterogeneity and Combining results from Several Studies in Meta-Analysis. Systematic Reviews in Health Care, John Wiley & Sons, Ltd 2001:285–312. https://doi.org/10.1002/9780470693926.ch15.
[21] Higgins JPT, Thompson SG. Quantifying heterogeneity in a meta-analysis. Stat. Med. 2002;21:1539–58. https://doi.org/10.1002/sim.1186.
[22] Borenstein M, Hedges LV, Higgins JPT, Rothstein HR. Introduction to Meta-Analysis. John Wiley & Sons 2009.
[23] Egger M, Smith GD, Schneider M, Minder C. Bias in meta-analysis detected by a simple, graphical test. BMJ 1997;315:629–34. https://doi.org/10.1136/bmj.315.7109.629.
[24] Sterne JAC, Egger M. Funnel plots for detecting bias in meta-analysis: guidelines on choice of axis. J. Clin. Epidemiol. 2001;54:1046–55. https://doi.org/10.1016/S0895-4356(01)00377-8.
[25] Viechtbauer W, Cheung M-W-L. Outlier and influence diagnostics for meta-analysis. Res. Synth. Methods 2010;1:112–25. https://doi.org/10.1002/jrsm.11.
[26] Henderson D, Zafar A, Bjornson A, Razak A, Achawal S, Danciut M, et al. Prognostic factors following resection of intracranial metastases. Surg. Neurol. Int. 2022;13:219. https://doi.org/10.25259/SNI_103_2022.
[27] Aftahy AK, Barz M, Lange N, Baumgart L, Thunstedt C, Eller MA, et al. The Impact of Postoperative Tumor Burden on patients with Brain Metastases. Front. Oncol. 2022;12:869764. https://doi.org/10.3389/fonc.2022.869764.
[28] Sander C, Frydrychowicz C, Prasse G, Taubenheim S, Arlt F, Meixensberger J, et al. The impact of neurological performance and volumetrics on overall survival in

brain metastasis in colorectal cancer: a retrospective single-center case series. BMC Cancer 2022;22:336. https://doi.org/10.1186/s12885-022-09435-1.
[29] Winther RR, Hjermstad MJ, Skovlund E, Aass N, Helseth E, Kaasa S, et al. Surgery for brain metastases—impact of the extent of resection. Acta Neurochir. 2022;164: 2773–80. https://doi.org/10.1007/s00701-021-05104-7.
[30] Koo J, Roh TH, Lee SR, Heo J, Oh Y-T, Kim S-H. Whole-Brain Radiotherapy vs. Localized Radiotherapy after Resection of Brain Metastases in the Era of Targeted Therapy: a Retrospective Study. Cancers 2021;13:4711. https://doi.org/10.3390/cancers13184711.
[31] Prabhu RS, Dhakal R, Vaslow ZK, Dan T, Mishra MV, Murphy ES, et al. Preoperative Radiosurgery for Resected Brain Metastases: the PROPS-BM Multicenter Cohort Study. International Journal of Radiation Oncology*Biology*Physics 2021;111:764–72. https://doi.org/10.1016/j.ijrobp.2021.05.124.
[32] Jünger ST, Pennig L, Schödel P, Goldbrunner R, Friker L, Kocher M, et al. The Debatable Benefit of Gross-Total Resection of Brain Metastases in a Comprehensive Treatment setting. Cancers 2021;13:1435. https://doi.org/10.3390/cancers13061435.
[33] McHugh FA, Kow CY, Falkov A, Heppner P, Law A, Bok A, et al. Metastatic melanoma: Surgical treatment of brain metastases – Analysis of 110 patients. J. Clin. Neurosci. 2020;73:144–9. https://doi.org/10.1016/j.jocn.2019.12.063.
[34] Olesrud IC, Schulz MK, Marcovic L, Kristensen BW, Pedersen CB, Kristiansen C, et al. Early postoperative MRI after resection of brain metastases—complete tumour resection associated with prolonged survival. Acta Neurochir. 2019;161: 555–65. https://doi.org/10.1007/s00701-019-03829-0.
[35] Calluaud G, Terrier L-M, Mathon B, Destrieux C, Velut S, François P, et al. Peritumoral Edema/Tumor volume Ratio: a strong Survival Predictor for Posterior Fossa Metastases. Neurosurg 2019;85:117–25. https://doi.org/10.1093/neuros/nyy222.
[36] She C, Wang R, Lu C, Sun Z, Li P, Yin Q, et al. Prognostic factors and outcome of surgically treated patients with brain metastases of non-small cell lung cancer. Thoracic Cancer 2019;10:137–42. https://doi.org/10.1111/1759-7714.12913.
[37] Sivasanker M, Madhugiri VS, Moiyadi AV, Shetty P, Subi TS. Surgery for brain metastases: an analysis of outcomes and factors affecting survival. Clin. Neurol. Neurosurg. 2018;168:153–62. https://doi.org/10.1016/j.clineuro.2018.03.011.
[38] Kerschbaumer J, Bauer M, Popovscaia M, Grams AE, Thomé C, Freyschlag CF. Correlation of Tumor and Peritumoral Edema Volumes with Survival in patients with Cerebral Metastases. AR 2017;37:871–6. https://doi.org/10.21873/anticanres.11392.
[39] Rava P, Rosenberg J, Jamorabo D, Sioshansi S, DiPetrillo T, Wazer DE, et al. Feasibility and safety of cavity-directed stereotactic radiosurgery for brain metastases at a high-volume medical center. Adv. Radiat. Oncol. 2016;1:141–7. https://doi.org/10.1016/j.adro.2016.06.002.
[40] Gazzeri R, Nalavenkata S, Teo C. Minimally invasive key-hole approach for the surgical treatment of single and multiple brain metastases. Clin. Neurol. Neurosurg. 2014;123:117–26. https://doi.org/10.1016/j.clineuro.2014.05.010.
[41] Lee C-H, Kim DG, Kim JW, Han JH, Kim YH, Park C-K, et al. The role of surgical resection in the management of brain metastasis: a 17-year longitudinal study. Acta Neurochir. 2013;155:389–97. https://doi.org/10.1007/s00701-013-1619-y.
[42] Auslands K, Apškalne D, Bicāns K, Ozols R, Ozoli H. Postoperative Survival in patients with Multiple Brain Metastases. Medicina 2012;48:41. https://doi.org/10.3390/medicina48060041.
[43] Limbrick DD, Lusis EA, Chicoine MR, Rich KM, Dacey RG, Dowling JL, et al. Combined surgical resection and stereotactic radiosurgery for treatment of cerebral metastases. Surg. Neurol. 2009;71:280–8. https://doi.org/10.1016/j.surneu.2007.12.019.
[44] Rades D, Pluemer A, Veninga T, Dunst J, Schild SE. A boost in addition to whole-brain radiotherapy improves patient outcome after resection of 1 or 2 brain metastases in recursive partitioning analysis class 1 and 2 patients. Cancer 2007; 110:1551–9. https://doi.org/10.1002/cncr.22960.
[45] Sauvageot S, Mollevi C, Thomas QD, Charissoux M, Darlix A, Rigau V, et al. Prognostic impact of the number and total tumor burden of secondary cerebral lesions in patients with resected brain metastases of non–small cell lung cancers. J. Neurosurg. 2024:1–11. https://doi.org/10.3171/2023.11.JNS231923.
[46] Baumgart L, Aftahy AK, Anetsberger A, Thunstedt D, Wiestler B, Bernhardt D, et al. Brain metastases in the elderly - Impact of residual tumor volume on overall survival. Front. Oncol. 2023;13:1149628. https://doi.org/10.3389/fonc.2023.1149628.
[47] Grossenbacher B, Lareida A, Moors S, Roth P, Kulcsar Z, Regli L, et al. Prognostic assessment in patients operated for brain metastasis from systemic tumors. Cancer Med. 2023;12:12316–24. https://doi.org/10.1002/cam4.5928.
[48] Shah PP, Franke JL, Medikonda R, Jackson CM, Srivastava S, Choi J, et al. Mutation status and postresection survival of patients with non–small cell lung cancer brain metastasis: implications of biomarker-driven therapy. J. Neurosurg. 2021;136:56–66. https://doi.org/10.3171/2020.10.JNS201787.
[49] Jünger ST, Reinecke D, Meissner A-K, Goldbrunner R, Grau S. Resection of symptomatic non–small cell lung cancer brain metastasis in the setting of multiple brain metastases. J. Neurosurg. 2021;136:1576–82. https://doi.org/10.3171/2021.7.JNS211172.
[50] Eitz KA, Lo SS, Soliman H, Sahgal A, Theriault A, MarkB P, et al. Multi-institutional Analysis of Prognostic Factors and Outcomes after Hypofractionated Stereotactic Radiotherapy to the Resection Cavity in patients with Brain Metastases. JAMA Oncol. 2020;6:1901–9. https://doi.org/10.1001/jamaoncol.2020.4630.
[51] Schödel P, Jünger ST, Wittersheim M, Reinhardt HC, Schmidt N-O, Goldbrunner R, et al. Surgical resection of symptomatic brain metastases improves the clinical status and facilitates further treatment. Cancer Med. 2020;9:7503–10. https://doi.org/10.1002/cam4.3402.
[52] Antuña AR, Vega MA, Sanchez CR, Fernandez VM. Brain Metastases of Non–Small Cell Lung Cancer: Prognostic Factors in patients with Surgical Resection. J Neurol Surg A Cent Eur Neurosurg 2018;79:101–7. https://doi.org/10.1055/s-0037-1601874.
[53] Kamp MA, Fischer I, Dibué-Adjei M, Munoz-Bendix C, Cornelius J-F, Steiger H-J, et al. Predictors for a further local in-brain progression after re-craniotomy of locally recurrent cerebral metastases. Neurosurg. Rev. 2018;41:813–23. https://doi.org/10.1007/s10143-017-0931-z.
[54] Schackert G, Lindner C, Petschke S, Leimert M, Kirsch M. Retrospective study of 127 surgically treated patients with multiple brain metastases: indication, prognostic factors, and outcome. Acta Neurochir. 2013;155:379–87. https://doi.org/10.1007/s00701-012-1606-8.
[55] Schackert G, Schmiedel K, Lindner C, Leimert M, Kirsch M. Surgery of recurrent brain metastases: retrospective analysis of 67 patients. Acta Neurochir. 2013;155: 1823–32. https://doi.org/10.1007/s00701-013-1821-y.
[56] Hong N, Yoo H, Gwak HS, Shin SH, Lee SH. Outcome of Surgical Resection of Symptomatic Cerebral Lesions in Non-Small Cell Lung Cancer patients with Multiple Brain Metastases. Brain Tumor Res Treat 2013;1:64–70. https://doi.org/10.14791/btrt.2013.1.2.64.
[57] Schödel P, Schebesch K-M, Brawanski A, Proescholdt MA. Surgical Resection of Brain Metastases—Impact on Neurological Outcome. Int. J. Mol. Sci. 2013;14: 8708–18. https://doi.org/10.3390/ijms14058708.
[58] Yoo H, Kim YZ, Nam BH, Shin SH, Yang HS, Lee JS, et al. Reduced local recurrence of a single brain metastasis through microscopic total resection: Clinical article. J. Neurosurg. 2009;110:730–6. https://doi.org/10.3171/2008.8.JNS08448.
[59] Tendulkar RD, Liu SW, Barnett GH, Vogelbaum MA, Toms SA, Jin T, et al. RPA classification has prognostic significance for surgically resected single brain metastasis. International Journal of Radiation Oncology*Biology*Physics 2006;66: 810–7. https://doi.org/10.1016/j.ijrobp.2006.06.003.
[60] Paek SH, Audu PB, Sperling MR, Cho J, Andrews DW. Reevaluation of Surgery for the Treatment of Brain Metastases: Review of 208 patients with Single or Multiple Brain Metastases Treated at one Institution with Modern Neurosurgical Techniques. Neurosurgery 2005;56:1021–34. https://doi.org/10.1227/01.NEU.0000158321.90608.BE.
[61] Stark AM, Tscheslog H, Buhl R, Held-Feindt J, Mehdorn HM. Surgical treatment for brain metastases: prognostic factors and survival in 177 patients. Neurosurg. Rev. 2005;28:115–9. https://doi.org/10.1007/s10143-004-0364-3.
[62] Bindal RK, Sawaya R, Leavens ME, Lee JJ. Surgical treatment of multiple brain metastases. J. Neurosurg. 1993;79:210–6. https://doi.org/10.3171/jns.1993.79.2.0210.
[63] Vogelbaum MA, Brown PD, Messersmith H, Brastianos PK, Burri S, Cahill D, et al. Treatment for Brain Metastases: ASCO-SNO-ASTRO Guideline. JCO 2021;40: 492–516. https://doi.org/10.1200/JCO.21.02314.
[64] Bellomo J, Zeitlberger AM, Padevit L, Stumpo V, Gönel M, Fierstra J, et al. Role of microsurgical tumor burden reduction in patients with breast cancer brain metastases considering molecular subtypes: a two-center volumetric survival analysis. J. Neurooncol 2024;169:379–90. https://doi.org/10.1007/s11060-024-04728-w.
[65] Wang J, Tawbi HA. Emergent immunotherapy approaches for brain metastases. Neuro-Oncol. Adv. 2021;3:v43–51. https://doi.org/10.1093/noajnl/vdab138.
[66] Tawbi HA, Forsyth PA, Algazi A, Hamid O, Hodi FS, Moschos SJ, et al. Combined Nivolumab and Ipilimumab in Melanoma Metastatic to the Brain. N. Engl. J. Med. 2018;379:722–30. https://doi.org/10.1056/NEJMoa1805453.
[67] Tawbi HA, Forsyth PA, Hodi FS, Algazi AP, Hamid O, Lao CD, et al. Long-term outcomes of patients with active melanoma brain metastases treated with combination nivolumab plus ipilimumab (CheckMate 204): final results of an open-label, multicentre, phase 2 study. Lancet Oncol. 2021;22:1692–704. https://doi.org/10.1016/S1470-2045(21)00545-3.
[68] Mahajan A, Ahmed S, McAleer MF, Weinberg JS, Li J, Brown P, et al. Post-operative stereotactic radiosurgery versus observation for completely resected brain metastases: a single-centre, randomised, controlled, phase 3 trial. Lancet Oncol. 2017;18:1040–8. https://doi.org/10.1016/S1470-2045(17)30414-X.
[69] Kochanek C, Gilde C, Zimmer L, Ugurel S, Meier F, Utikal J, et al. Effects of an immunosuppressive therapy on the efficacy of immune checkpoint inhibition in metastatic melanoma – an analysis of the prospective skin cancer registry ADOREG. Eur. J. Cancer 2024;198:113508. https://doi.org/10.1016/j.ejca.2023.113508.
[70] Jessurun CAC, Hulsbergen AFC, de Wit AE, Tewarie IA, Snijders TJ, Verhoeff JJC, et al. The combined use of steroids and immune checkpoint inhibitors in brain metastasis patients: a systematic review and meta-analysis. Neuro Oncol. 2021;23: 1261–72. https://doi.org/10.1093/neuonc/noab046.
[71] Alvarez-Breckenridge C, Giobbie-Hurder A, Gill CM, Bertalan M, Stocking J, Kaplan A, et al. Upfront Surgical Resection of Melanoma Brain Metastases Provides a Bridge Toward Immunotherapy-Mediated Systemic Control. Oncologist 2019;24: 671–9. https://doi.org/10.1634/theoncologist.2018-0306.
[72] Dall'Olio FG, Marabelle A, Caramella C, Garcia C, Aldea M, Chaput N,, et al. Tumour burden and efficacy of immune-checkpoint inhibitors. Nat. Rev. Clin. Oncol. 2022;19:75–90. https://doi.org/10.1038/s41571-021-00564-3.
[73] Pollock BE, Brown PD, Foote RL, Stafford SL, Schomberg PJ. Properly selected patients with Multiple Brain Metastases May Benefit from Aggressive Treatment of their Intracranial Disease. J. Neurooncol 2003;61:73–80. https://doi.org/10.1023/A:1021262218151.
[74] Tanei T, Fujii M, Takebayashi S, Nakahara N, Wakabayashi T. Simultaneous multiple craniotomies in the management of multifocal malignant brain lesions:

case reports. Fukushima J. Med. Sci. 2019;65:43–9. https://doi.org/10.5387/fms.2019-06.

[75] Bschorer M, Ricklefs FL, Sauvigny T, Westphal M, Dührsen L. Multiple craniotomies in a single surgery — the resection of scattered brain metastases. Neurosurg. Rev. 2023;46:70. https://doi.org/10.1007/s10143-023-01976-8.

[76] Niedermeyer S, Schmutzer-Sondergeld M, Weller J, Katzendobler S, Kirchleitner S, Forbrig R, et al. Neurosurgical resection of multiple brain metastases: outcomes, complications, and survival rates in a retrospective analysis. J. Neurooncol 2024;169:349–58. https://doi.org/10.1007/s11060-024-04744-w.

[77] Padevit L, Zeitlberger AM, Maldaner N, Sarnthein J, Bozinov O, Regli L, et al. Multiple craniotomies in patients with brain metastases: a two-center, propensity score-matched study. Neurosurg. Rev. 2024;47:354. https://doi.org/10.1007/s10143-024-02578-8.

[78] Kotecha R, La Rosa A, Brown PD, Vogelbaum MA, Navarria P, Bodensohn R, et al. Multidisciplinary management strategies for recurrent brain metastasis after prior radiotherapy: An overview. Neuro-Oncology 2024:noae220. https://doi.org/10.1093/neuonc/noae220.

[79] Das S, Faruqi S, Nordal R, Starreveld Y, Kelly J, Bowden G, et al. A phase III, multicenter, randomized controlled trial of preoperative versus postoperative stereotactic radiosurgery for patients with surgically resectable brain metastases. BMC Cancer 2022;22:1368. https://doi.org/10.1186/s12885-022-10480-z.

[80] Rogers S, Schwyzer L, Lomax N, Alonso S, Lazeroms T, Gomez S, et al. Preoperative radiosurgery for brain metastases (PREOP-1): A feasibility trial. ctRO 2024;47. https://doi.org/10.1016/j.ctro.2024.100798.

[81] Agrawal N, Shireman JM, Shiue K, Kamer A, Boyd L, Zang Y, et al. Preoperative stereotactic radiosurgery for patients with 1–4 brain metastases: a single-arm phase 2 trial outcome analysis (NCT03398694). Neuro-Oncol. Pract. 2024;11:593–603. https://doi.org/10.1093/nop/npae043.

[82] Mallela AN, Iheagwara UK, Fogg D, Anthony A, Gersey ZC, Zhang X, et al. Preoperative stereotactic radiosurgery for cerebral metastases: safe, effective, and decreases steroid dependency. J. Neurosurg. 2024;141:1332–42. https://doi.org/10.3171/2024.2.JNS232426.

[83] Brastianos PK, Carter SL, Santagata S, Cahill DP, Taylor-Weiner A, Jones RT, et al. Genomic Characterization of Brain Metastases reveals Branched Evolution and potential Therapeutic Targets. Cancer Discov. 2015;5:1164–77. https://doi.org/10.1158/2159-8290.CD-15-0369.